# SAINT MICHAEL COLLEGE OF CARAGA

# WEB-BASED SMART LEAP APPLICATION AND REMITTANCE SYSTEM FOR CSWDD BUTUAN CITY

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TABLE OF CONTENTS

Pages
TITLE PAGE i
CERTIFICATE OF RESEARCH APPROVAL ii
ABSTRACT iii
DEDICATION iv
ACKNOWLEDGEMENT v
TABLE OF CONTENTS vi
LIST OF TABLES viii
LIST OF FIGURES x

CHAPTER
1 **INTRODUCTION** 1
1.1 Project Context 1
1.2 Literature Review 5
1.3 Objective of the Study 13
1.4 Scope and Limitations 14
1.5 Definition of Terms 14

2 **METHODOLOGY** 18
2.1 Research Design 18
2.2 Software Development Life Cycle 18
2.3 System Architecture 19
2.4 Conceptual Diagram 21
2.5 Use Case Diagram 23
2.6 Activity Diagram 25
2.7 Sequence Diagram 31
2.8 Entity-Relationship Diagram (Database Design) 41
2.9 User Interface Design (Prototype) 43
2.10 Software Platforms, Development Environments, and Tools 68
2.11 Hardware Requirements 69
2.12 Research Ethical Consideration 70
2.13 System Evaluation 71
2.13.1 Functional Suitability 71
2.13.2 Performance Efficiency 71
2.13.3 Usability 71

REFERENCES 73
SAMPLE ETHICAL CONSIDERATIONS 77
CURRICULUM VITAE 81

## LIST OF TABLES

| Table | | Pages |
|---|---|---|
| 1 | Software Platforms, Development Environments, and Tools | 68 |
| 2 | Hardware Requirements | 69 |
| 3 | Functional Suitability | 70 |
| 4 | Performance Efficiency | 71 |
| 5 | Usability | 71 |

## LIST OF FIGURES

| Figure | | Pages |
|---|---|---|
| 1 | Agile Methodology | 18 |
| 2 | System Architecture | 19 |
| 3 | Conceptual Framework | 21 |
| 4 | Use Case Diagram | 23 |
| 5 | Activity Diagram (Beneficiary) | 25 |
| 6 | Activity Diagram (PDO) | 27 |
| 7 | Activity Diagram (Admin) | 29 |
| 8 | Activity Diagram (Social Worker) | 30 |
| 9 | Sequence Diagram (Admin) | 32 |
| 10 | Sequence Diagram (PDO) | 34 |
| 11 | Sequence Diagram (Social Worker) | 37 |
| 12 | Sequence Diagram (Beneficiary) | 39 |
| 13 | Entity Relationship Diagram | 40 |
| 14 | Login Page (Admin, PDO, Social Worker) | 42 |
| 15 | Admin Dashboard | 44 |
| 16 | Admin & PDO Application Management | 46 |
| 17 | Repayments (Admin & PDO) | 47 |
| 18 | Training (Admin & PDO) | 49 |
| 19 | Reports & Analytics (Admin) | 51 |
| 20 | User Management (Admin) | 52 |

| | | |
|---|---|---|
| 21 | PDO Dashboard | 53 |
| 22 | Social Worker Dashboard | 54 |
| 23 | Beneficiary Portal | 56 |
| 24 | Beneficiary Application Forms | 61 |
| 25 | Beneficiary Dashboard | 64 |

# CHAPTER 1

# INTRODUCTION

## 1.1 Project Context

The City Social Welfare and Development Department (CSWDD) operates through the Local Government Unit (LGU) of Butuan City. Its main job is to promote social protection and inclusive growth. The CSWDD's role is to develop and implement programs that support individuals, families, and groups in need, particularly those facing financial or social challenges. The Sustainable Market and Technology-Driven Livelihood and Employment Assistance Program, or SMART LEAP, is one of its most important projects. Low-income people can start and run small businesses with the help of this program. It provides them with money, valuable tools, and training they can use immediately.

The primary goal of SMART LEAP is to equip individuals with the necessary tools to enhance their lives through entrepreneurship and empower them to become self-sufficient. It does more than give people money; it also helps them improve in various areas and provides ongoing guidance to support their success. The program's primary goal is to assist groups that are not receiving sufficient support, and it continually evolves to adapt to new tools and market demands. A strong approach to combat poverty and stimulate the economy in Butuan City is essential for its growth and development. SMART LEAP has a significant goal, but it must address real-world problems.

The current system still relies heavily on manual tasks, which slow down the process and increase the time required to verify recipients, track repayments, and generate reports. When staff members handle paperwork on paper, they commonly make mistakes and often lose records, which makes it challenging to track progress or measure effects. Transparency is lost when there is no single digital database or an easy way for beneficiaries to voice their concerns. People who live in remote places have an even harder time getting services or help.

Many people are interested in updating social safety programs by digitizing them. Governments worldwide are striving to enhance their aid delivery. Lowe [1] said that many social safety agencies still use disconnected or only partially digital registries. This results in incorrect data, overly time-consuming verification processes, and inadequate tracking capabilities. Her research revealed that these inefficiencies hinder agencies' ability to track beneficiaries and review outcomes effectively and accurately.

These problems persist in areas that have only partially adopted digital platforms, where systems often fail to integrate or lack effective data-sharing methods. Zulkifli et al. [2] also looked at how government-linked groups are going digital and found that switching from paper to electronic records can be hard if the right policies, equipment, and training are not in place. Businesses could lose data, make it difficult to access, or receive insufficient data if they do not receive the proper assistance. The study concluded that adopting digital technology requires more than just new technology. It also requires well-equipped institutions, effective leadership, and skilled staff.

Local government units (LGUs) primarily carry out social aid programs and e-governance projects in the Philippines. Camorongan [3] found that many LGUs struggle with poorly integrated ICT infrastructures, inconsistent use of digital systems, and poor coordination among agencies. Although national guidelines support smart-city projects and digital platforms, local implementation often faces challenges due to inadequate resources, staff training, and inconsistent data management frameworks. These issues make it challenging for local governments to act promptly, maintain transparency, and effectively engage citizens in public services. The study highlights that people often encounter broken or missing channels to communicate with the government. This lack of integration leads to reduced accountability and delays in resolving issues. Many local governments rely on paper records and separate, unintegrated reporting systems, making it challenging to align data or access results promptly. These circumstances demonstrate the ongoing struggles.

Bigger problems with the growth of e-governance in the country, as digital infrastructure is still not uniform across all areas. The SMART LEAP program from the CSWDD aims to help individuals find a market-focused career path that empowers them through technology. However, the continued use of manual processes undermines this goal. Handwritten records and physical submissions slow transactions and increase the risk of data handling mistakes, lost papers, and duplicate entries. Such errors make it harder to evaluate programs and prevent the department from helping people who need help at the right time. Additionally, program heads struggle to produce accurate reports and evaluate success indicators due to the absence of an electronic monitoring and tracking system. Hasty decision-making can lead to policy choices that lack careful thought. CSWDD Administrators, social workers, and project development officers do not use the same digital tools. The situation makes it more challenging for team members to communicate with each other and collaborate effectively, resulting in repeated tasks and late report submissions.

The researchers aim to address these issues by developing a Web-based SMART LEAP application and remittance system. The system will automate application submission, verify beneficiaries' eligibility, track training attendance, and send monthly payments. It will also enable beneficiaries to pay their payments digitally by uploading the official receipt from Butuan City Hall, reducing paperwork and transcription errors. A central database will enable all approved users, including project development officers, social workers, and administrators, to access up-to-date information.

Dashboards will be built into the proposed platform to make data easier to understand. Managers can quickly check the progress of programs and the company's finances with this tool or system. Built-in verification and notification systems will inform users about incomplete applications or funds that are past due, enabling them to make informed decisions and take prompt action. Secure login credentials and encrypted data storage will also help keep private data safe and ensure compliance with privacy and confidentiality standards. With this new feature, the system should make work easier, eliminate unnecessary tasks, and facilitate collaboration among people involved in the program. The following plan will serve as the basis for the CSWDD's future digital projects, helping it move closer to running its online business.

Implementing the proposed system is crucial for both the CSWDD and the overall effort to modernize the Philippine public administration. By going digital with the SMART LEAP program, the department can make its operations more transparent, reduce staff errors, and expedite service delivery to recipients. The national government is trying to become more digital through its e-governance and open-data programs. This new idea helps with that. The system will also help build trust in the government by making data more accurate and by holding administrators accountable for how they spend livelihood-assistance funds. Beneficiaries will receive faster processing and clearer communication, and managers will have access to real-time information to make informed decisions based on facts. The project will also serve as an example for other LGUs that wish to utilize locally available tools to digitize their welfare programs.

Despite the existence of livelihood assistance programs such as SMART LEAP, the lack of a unified, technology-driven system limits the program's effectiveness and sustainability. The absence of an integrated digital platform results in fragmented data management, delayed transactions, weak monitoring mechanisms, and limited accessibility for beneficiaries, particularly those in geographically isolated areas. These gaps highlight the need for a centralized, web-based solution that can streamline operations, improve transparency, and support evidence-based decision-making. Without addressing these limitations, the CSWDD may continue to face operational inefficiencies that hinder the timely delivery of services and the accurate evaluation of program outcomes.

## 1. 2 **Literature Review**

This section outlines the relevant literature that supports the researchers' examination of the present issues related to livelihood assistance repayment and oversight. Thematic analysis of digitization mainly concentrates on efficiency, transparency, and equity. Critical perspectives caution against overreliance on digital platforms, highlighting concerns such as unintended consequences, infrastructural limitations, and challenges related to digital literacy. This balanced theme structure emphasizes the importance of secure, inclusive, and well-designed digital solutions, while also situating the current study within the broader academic conversation.

### ***Digital Payments from Government to People and Social Protection***

Digitalizing Government-to-Person (G2P) systems is commonly seen as a big step forward in how the government helps people. Theis et al. [4] examined Indonesia's Keluarga Harapan (PKH) program, a conditional cash-transfer initiative that transitioned from manual to digital disbursement of funds. They sought to know how digital channels affected women's ability to be included and empowered. The analysis shows that digitization has made payments more timely and clear. However, delays and exclusions continue to occur due to manual verification, poor data integration, and recipients' limited digital skills. It concluded that digitalization alone cannot guarantee integration without the incorporation of interoperable technologies and user-centered program design.

Niankara and Islam [5] examined the impact of B2P and G2P digital transfers on financial inclusion at a macroeconomic level, employing models from several nations. Their research demonstrated that electronic transfers significantly augmented the prevalence of ownership and savings accounts. However, these benefits relied on a robust digital infrastructure and inter-institutional collaboration. Vujović et al. [6] argue that research conducted in South Africa suggests that digital social protection systems enhance transparency and equity. However, administrative silos and a lack of cooperation across agencies still hinder efficiency. The authors suggested that combining social registrations with real-time data analytics could address these management issues.

Kangasniemi et al. [7] also looked at how policies that protect people could help them be more accepting and strong. They argued that digital systems need to have strong ways to keep an eye on things to make sure that welfare programs serve individuals on the edge. Research in El Salvador [8] demonstrated that digital remittances exert analogous effects. It was decided that electronic transfers make it easier for people to acquire money, are less costly to conduct business, and generate revenue more quickly. However, they need to be monitored to ensure they are used correctly and to maintain people's privacy.

These studies demonstrate that transitioning to digital in G2P programs enhances speed, transparency, and participation. However, its full potential is often limited by broken systems, manual checks, and poor data integration. The results clearly indicate that digitalization should encompass more than just online payments. It should also include interoperable registries, automatic verification, and ways for people to give feedback. For local programs like the SMART LEAP program to be effective, they must utilize a digital workflow that integrates registration, verification, payment, and tracking into a single platform. This type of integration ensures that everyone is accountable, reduces unnecessary work, and promotes a fair distribution of livelihood aid.

The literature study reveals that digital G2P systems are a crucial means an updating social protection. However, they require well-coordinated databases, collaboration between institutions, and a design that focuses on the user to function effectively. Indonesia and El Salvador have become more open and efficient, but they still face similar problems to those addressed in the SMART LEAP program run by the CSWDD. Some of these problems include corrupted data, slow verification, and restricted access for individuals unfamiliar with computer use. To make SMART LEAP into a clear, compelling, and inclusive digital welfare model that meets global standards for digital government, we need a system that is fully integrated and automated.

## Public Sector Digitalization

Digital transformation in the public sector is a crucial aspect of good governance, particularly in developing countries seeking to modernize their administrative processes. Owolabi [9] examined the potential of blockchain in maintaining government records. He found that its decentralized nature makes it transparent, unchangeable, and traceable, which are all important qualities that stop document tampering and fraud. The report, on the other hand, stated that adopting blockchain requires clear rules, additional training, and long-term investments in digital infrastructure. Riley and Shonchoy [10] assessed Ghana's national campaign promoting fintech adoption and discovered that digital projects frequently falter in the absence of systematic data collection and well-coordinated communication methods. Their results show that to keep citizens interested, well-organized data management systems must support technical changes.

Senyo et al. [11] created a conceptual model of "government-led digital transformation" in fintech ecosystems. They said that leadership, coordination, and collaboration amongst agencies are what make public-sector innovation work. The research indicated that nations that synchronize their fintech ecosystems with governmental policy frameworks get more rapid and enduring innovation results. Ly [12] demonstrated empirically that digital transformation in public firms depends on leadership development and organizational learning. The study employed PLS-SEM analysis to establish a favorable relationship between leadership capability, digital strategy, and sustainability performance, highlighting that transformation necessitates both technological and cultural adaptation. Moreover, related research indicates that without strong governance, technology implementation is prone to stagnation. Aurazo et al. [13] studied central bank digital currencies (CBDCs) and found that interoperability and data governance affect how financial innovations change systems. Senyo et al. [11] and Owolabi [9] concur that the absence of a distinct policy direction frequently results in departments duplicating efforts, hence diminishing their potential efficiency.

From all these perspectives, scholars agree on one thing: digital transformation can only succeed if clear rules guide it and include everyone in the institutions that operate within it. Policy fragmentation, staff lack of training, and the absence of interoperable systems all hinder innovation and render upgrading efforts less effective. Aurazo et al. [13] added to the importance of interoperability and reasonable data control

In avoiding duplication and making sure that new technologies, like digital currencies issued by central banks, improve rather than complicate government work. The main idea that consistently emerges in these studies is that leadership, teamwork, and clear policy direction are key factors in determining whether digitalization enhances administrative efficiency.

In short, the research indicates that the digital transformation of the public sector is not merely a technical change; it is also a strategic shift in how things are managed. Strong leadership, coordinated data management, and clear policy frameworks that combine technology with institutional capabilities are all necessary for implementation to proceed smoothly. Blockchain and fintech are two innovations that hold great promise for making things more open and efficient. Even though new technologies like blockchain and fintech hold much promise to make things more open and efficient, they could still fail if they are not adequately governed and departments do not work together effectively.

## Digital Financial Inclusion

Digital financial inclusion (DFI) is gaining increasing popularity as a tool to help individuals move out of poverty and improve their lives. Flaming and Jeník [14] found that when microfinance institutions utilized digital tools such as e-KYC, mobile wallets, and agent networks, they improved their customer service. However, continuous investment in staff training and digital competencies was essential for successful implementation. Alom et al. [15] demonstrated that internet platforms offered women entrepreneurs in Bangladesh a genuine opportunity for success. These tools helped women build their enterprises and become financially independent by making it easier to access funds and reducing transaction costs.

Guan and Han [15] have stated that DFI benefits not only individuals but also strengthens entire regions. Their research in China demonstrated that digital access enhances business and education, therefore fostering economic growth. DFI also plays a small but important role in bringing people of all ages and from all over the world together. Cai et al. [16] found that older individuals, who are often excluded from conventional banking, can maintain financial engagement through digital platforms—whether for bill payment, seeking assistance, or managing small businesses. Huang et al. [17] stressed that DFI helps bring individuals from cities and towns closer together, giving more people a fair opportunity at financial security. However, Nyakurukwa et al. [18] and Osuma et al. [19] remind us that technology is not adequate on its own. For digital inclusion to function effectively in African countries, there must be robust infrastructure, digital literacy, and clear policies in place to protect users.

These studies show that DFI is both a technological and a developmental tool when looked at as a whole. It can improve people's lives and the economies of entire countries if it is implemented with inclusive planning, reliable infrastructure, and the development of people's skills. However, its benefits will not be shared equally without good governance, coordination, and ongoing teaching.

To summarize, digital financial inclusion is a transformative approach to helping people escape poverty and gain greater control, particularly in developing countries. The primary benefit is that DFI facilitates access to loans, supports the growth of small businesses, and encourages individuals to join the digital economy. Still, the key to success is combining technology with people's skills and fair rules.

**FinTech Adoption**

The rapid growth of fintech apps has transformed the way money works, but it has also raised new concerns about security and control. Pueblos and Timoteo [20] found that micro-entrepreneurs in Taguig City using e-payment systems achieved enhanced operational and sales efficiency; however, challenges related to reliability, infrastructure, and access remained. Bhutto et al. [21] conducted a study on HR Competency and Fintech Adoption in the banking sector, identifying service innovation and employee competence as significant drivers of successful adoption.

Başar et al. [22] examined 45 countries and validated that digital financial literacy has a substantial impact on saving behavior and the secure utilization of fintech. Offiong and SzopikDepczyńska [23] took a close look at how fintech is shaping microfinance, and their findings tell a story of both progress and caution. On one hand, going digital has helped microfinance institutions reach more people and operate more efficiently—making it easier for clients to access services and for providers to manage them. The research also raised a significant concern: as more financial activity moves online, users are increasingly vulnerable to new risks, such as data abuse and cyber threats. The report also raised a significant concern:

as more financial transactions occur online, people are at risk of incidents such as data theft and cyberattacks.

All of these studies concur on one point: strong governance frameworks, ongoing user education, ethical oversight, and secure digital environments are crucial for fintech to realize its full potential in terms of inclusion, transparency, and operational excellence. Without these protections, going digital could exacerbate inequality and leave users vulnerable to structural flaws, rather than empowering them.

In conclusion, the study demonstrates that utilizing fintech facilitates smoother operations, engages a broader audience, and generates innovative ideas. To make this work, however, people must be proficient with technology and possess strong moral principles. Digital payments and online platforms give small businesses more power and make it easier for more people to get money. However, they also put users at risk of cyberattacks and data privacy breaches. For this reason, for fintech systems to work, they need more than just cutting-edge technology. They also need clear rules, a strong infrastructure for protection, and ongoing training for both providers and users.

According to the collective body of literature, transitioning to digital platforms in the financial and public sectors is now necessary to make government and social development more efficient, transparent, and inclusive for all. Studies show that automating digital G2P payments improves the timeliness and accountability of aid distribution. However, issues such as manually verifying information, incorrectly combining data, and recipients not knowing how to use computers still render it less valuable. This type of research suggests that for social safety programs to be effective, everything must be conducted online, from registration to receiving the funds. As part of this, they will be watching and getting comments.

Research on the digitalization of the public sector reveals that technology alone does not guarantee modernization. Instead, transformation works best when it is supported by strong leadership, effective coordination, and cooperation among institutions. Key factors that determine lasting innovation include the ability of leaders, the consistency of policies, and the capacity of agencies to collaborate effectively. Similarly, research on digital financial inclusion emphasizes that digital tools can facilitate access to credit, encourage entrepreneurship, and reduce inequality, provided they are utilized in conjunction with robust infrastructure, user education, and inclusive programs. Without these digital divides will persist, and the most vulnerable individuals will be unable to benefit from modernization fully.

Lastly, examples from the use of fintech and payment security illustrate both the benefits and drawbacks of new financial technologies. While fintech systems make things more efficient and open up access to more people, they also raise concerns about cybersecurity, data privacy, and fair government. Experts warn that when digital finance is used, users must be protected, staff must be skilled, and appropriate oversight must be in place to keep the goal on track and protect public trust.

This list of four ideas demonstrates that going digital is not just a technological change; it is also a shift in how things are done. It is essential to strike a balance between new tools and people's existing skills, enabling innovation to enhance the fairness and accountability of institutions.

## 1.3 Objective of the Study

To design and develop a Web-Based SMART LEAP Assistance and Remittance System for the City Social Welfare and Development Department (CSWDD) of Butuan City that will digitize the online application, monitoring, verification, and tracking of beneficiaries' assistance repayments and training, thereby improving efficiency, transparency, and accountability in the program.

Specifically, the study aims to:

1. Automate the assistance application verification, evaluation, and approval process through integrated digital records management and validation tools.
2. Develop a digital remittance module that enables beneficiaries to upload proof of payment or receipts to verify partial and complete assistance repayments.
3. Create a training tracking module that records beneficiary attendance, progress, and completion in livelihood and entrepreneurship training programs under SMART LEAP.
4. Provide real-time monitoring and reporting features for CSWDD officers to oversee assistance status, training compliance, and overall beneficiary performance.
5. To evaluate the system based on the ISO/IEC 25010 Software Quality Model to ensure that it meets international standards in terms of functionality, usability, reliability, performance efficiency, security, maintainability, and portability.

## 1.4 Scope and Limitations

The SMART LEAP project aims to design and develop a web-based application for verification, monitoring, and repayment tracking systems for the City Social Welfare and Development Department (CSWDD) in Butuan City. This system replaces the manual processes of application intake and training tracking, remittance validation, and repayment monitoring with a centralized, database-driven solution that streamlines client registration, data organization, reporting, and feedback collection.

This program is exclusively for CSWDD personnel and SMART LEAP beneficiaries officially registered in the program database. The ICT Focal or an authorized staff member will run SMART LEAP using one admin account (superuser).End-users—the Project Development Officers, Social Workers, and Beneficiaries—will interact with the system through role-specific interfaces. The superuser will fully access all system features, including user management, client data control, Repayment and training tracking, and report generation. The system flow includes secure Google sign-in for the client. Registration, repayment schedule encoding, transaction logging, data visualization, and feedback collection for continuous improvement.

## 1.5. Definition of Terms

The definitions below clarify the fundamental concepts and elements addressed in this paper, ensuring clarity and uniformity in understanding essential processes, responsibilities, and system components included in the SMART-LEAP architecture.

**Assistance** - In the context of the SMART LEAP Assistance and Remittance System, "assistance" means the money that the City Social Welfare and Development Department (CSWDD) of Butuan City gives to eligible beneficiaries through the Sustainable Market and Technology-Driven Livelihood and

Employment Assistance Program (SMART LEAP). Typically, around ₱15,000, this funding serves as start-up capital for micro-enterprises or livelihood projects. The assistance provided is a loan rather than a grant, requiring Repayment. Recipients are obligated to repay ₱625 each month for 24 months.

**Backend** – The backend is the system's core infrastructure that operates behind the user interface. It is responsible for processing data, managing user authentication, enforcing access controls, and facilitating secure storage and retrieval of records. This component ensures the integrity, reliability, and capability of the SMART LEAP system, a database—oriented management system—a structured software system used to create, store, and manipulate databases. In the SMART LEAP Application and Remittance System, the DBMS manages records of beneficiaries, financial transactions, and assistance distribution, providing a secure and efficient way to maintain large amounts of sensitive data.

**End-user** - The individuals who utilize and interact with the system. In this context, end-users include CSWDD staff who encode and monitor assistance records and beneficiaries who apply for and track their cash assistance. Their access is limited to the functions necessary for their role.

**Frontend**- refers to the visible interface of the SMART LEAP system that users interact with directly. It encompasses all graphical elements such as forms, buttons, menus, and dashboards. This component allows beneficiaries and CSWDD personnel to navigate the system intuitively and responsively across various devices.

**Project Development Officer** - The City Social Welfare and Development Department (CSWDD) has a Project Development Officer (PDO) who is in charge of processing, reviewing, and keeping an eye on applications for and repayments of livelihood aid through the SMART LEAP aid and Remittance System. The PDO is responsible for the entire process, from profiling beneficiaries and verifying their eligibility to reviewing their documents and approving and releasing financial assistance. The PDO also ensures that individuals who receive assistance adhere to training and loan repayment rules.

**Receipt Upload** – A feature of the system that allows beneficiaries to submit a digital image of their official receipt (OR) issued by the City Treasurer's Office as proof of payment.

**Repayment Tracking** – We monitor the beneficiaries' monthly payment status (₱625 per month) and verify their partial and complete payments until they settle the entire loan.

**SMART LEAP** – Stands for Sustainable Market and Technology-driven Livelihood and Employment Assistance Program, a government initiative of the CSWDD in Butuan City that provides ₱15,000 financial assistance to beneficiaries, payable over 24 months with no interest.

**Statistical Report** – A highly privileged account for the SMART LEAP Application and Remittance System administration. In this case, the Superuser role refers to the ICT focal or system administrator of CSWDD Butuan City, who has full privileges to configure the system, manage user accounts, and oversee all transactions and reports.

**Superuser** – A highly privileged account for the SMART LEAP Application and Remittance System administration. In this case, the Superuser role refers to the ICT focal or system administrator of CSWDD Butuan City, who has full privileges to configure the system, manage user accounts, and oversee all transactions, trainings, and reports.

**Training** - refers to a planned learning program that gives beneficiaries the knowledge, skills, and attitudes they need to reach the program's development goals. Training sessions could include learning about finances, preparing for a job, becoming involved in the community, or getting acquainted with the program. Authorized staff plan and run each session, and we record them in the system to ensure accountability and traceability for everyone involved.

To be eligible for help, beneficiaries must attend and finish specific training courses. The Program Development Officer (PDO) checks to see if people have completed their work and keeps track of who has attended. Once validated, the system changes the Beneficiary's profile to indicate they are "Qualified," allowing them to proceed to the next step of receiving help.

# CHAPTER 2

# METHODOLOGY

## 2.1 Research Design

This study will use a Developmental Research Design (Systems Development Research). It focuses on designing, developing, and evaluating the Web-Based SMART LEAP Application and Remittance System for CSWDD, Butuan City. The research will follow the Design and Development (D&D) method, ensuring the system's functionality and usability.

**2.2 Software Development Life Cycle**

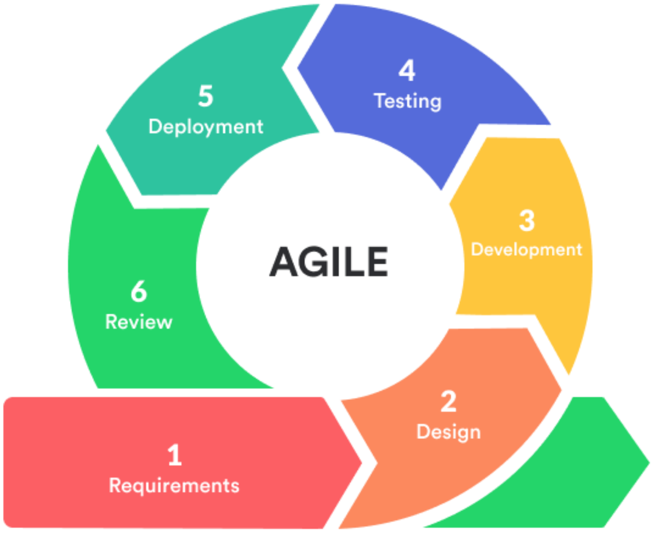


*Figure 1. Agile Methodology*

Figure 1 illustrates the Agile Methodology that suits the system well, as it promotes flexibility, cooperation, and continuous development—essential qualities for the Web-Based SMART LEAP Application and Remittance System for CSWDD Butuan City. Through agile, we quickly take action on changes and user feedback, so we are not reacting to system improvements. This approach provides the opportunity to adjust plans and features while the project runs, unlike traditional methods that only work once. Small steps are taken in each cycle to address the needs and problems immediately before they can grow. The results can also be delivered more quickly, allowing the team and stakeholders to see progress more easily. In this way, admin, PDOs, and social workers can collaborate more easily, ensuring the whole team is on the same page throughout the process. In other words, Agile is a way to speed up the job, be more efficient, and resolve issues faster, making it a good fit for the system.

## 2.3 System Architecture

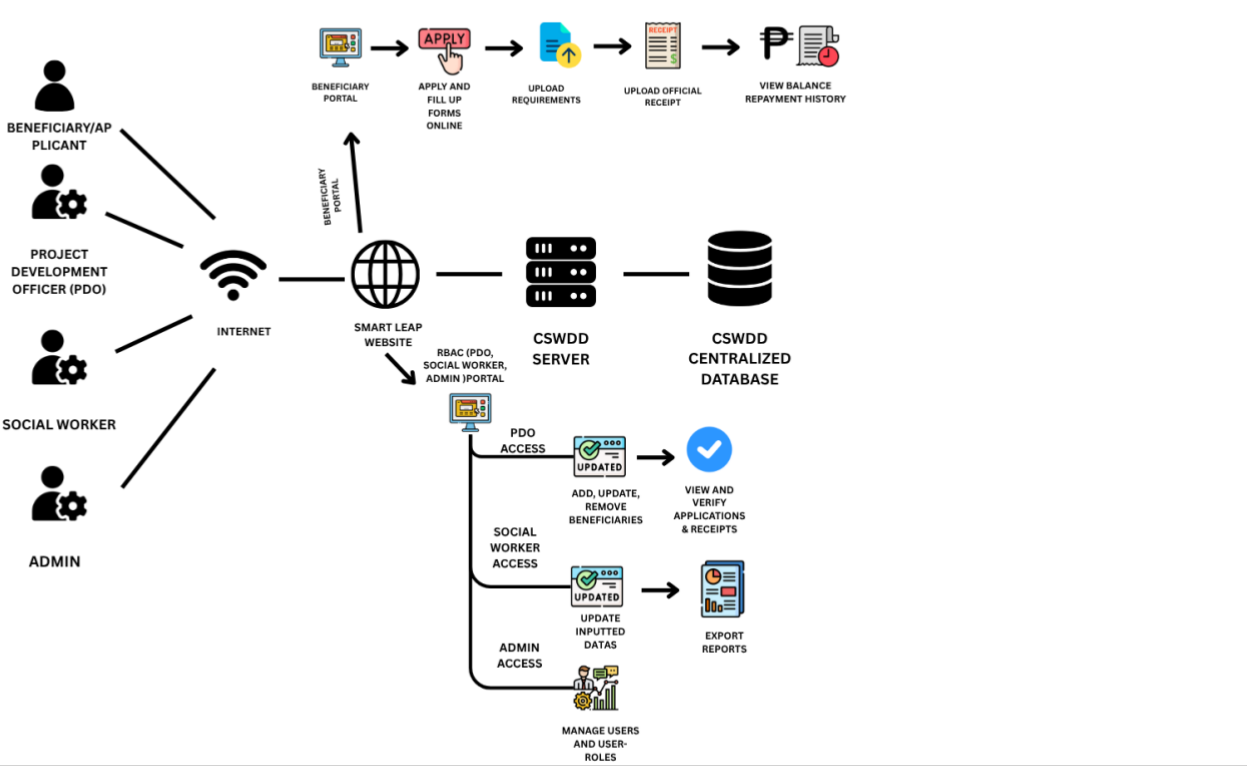

BENEFICIARY PORTAL
APPLY
APPLY AND FILL UP FORMS ONLINE
UPLOAD REQUIREMENTS
RECEIPT
UPLOAD OFFICIAL RECEIPT
₱
VIEW BALANCE REPAYMENT HISTORY
BENEFICIARY/APPLICANT
BENEFICIARY PORTAL
PROJECT DEVELOPMENT OFFICER (PDO)
INTERNET
SMART LEAP WEBSITE
CSWDD SERVER
CSWDD CENTRALIZED DATABASE
RBAC (PDO, SOCIAL WORKER, ADMIN )PORTAL
SOCIAL WORKER
PDO ACCESS
UPDATED
ADD, UPDATE, REMOVE BENEFICIARIES
VIEW AND VERIFY APPLICATIONS & RECEIPTS
ADMIN
SOCIAL WORKER ACCESS
UPDATED
UPDATE INPUTTED DATAS
EXPORT REPORTS
ADMIN ACCESS
MANAGE USERS AND USER-ROLES

*Figure 2. System Architecture*

Figure 2 illustrates the system architecture of the Web-Based SMART LEAP Application and Remittance System for CSWDD Butuan City, depicting how various users, portals, and procedures interact with one another through a centralized digital infrastructure. Beneficiaries, Project Development Officers (PDOs), social workers, and administrators are among the users who access the SMART LEAP web application online through the client-server architecture. Each user group operates within a dedicated portal that caters to their unique roles and responsibilities.

The Beneficiary or Applicant uses the Beneficiary Portal to apply online for livelihood assistance at the user level. Users can monitor their loan amount and repayment history in real-time, upload necessary documents, and submit formal payment receipts through this interface. The Project Development Officer (PDO) uses the PDO Portal to add, update, and validate applications and receipts and manage beneficiary records. Social workers can update the data they entered and export reports for monitoring and assessment through the Social Worker Portal. System administrator utilize the Admin Portal to control user administration, system configuration, and access privileges. Implementing this system enables all parties involved in CSWDD to coordinate securely and systematically.

The SMART LEAP Website handles all user interactions and data transactions. It connects to the CSWDD Server to validate and synchronize real-time data. The server serves as the primary mediator between user requests and the CSWDD Centralized Database, which safely stores all beneficiary profiles, remittance data, and report files. Data accessibility, accuracy, and consistency are guaranteed across all portals by the centralized database.

This system architecture fosters an integrated and smooth digital ecosystem where information moves effectively between users and the central repository. The SMART LEAP system enables CSWDD Butuan City to track repayments, monitor livelihood support programs, and manage digital transactions in an organized, transparent, and user-friendly manner by combining these elements into a single, secure platform.

**2.4 Conceptual Diagram (IPO Diagram)**

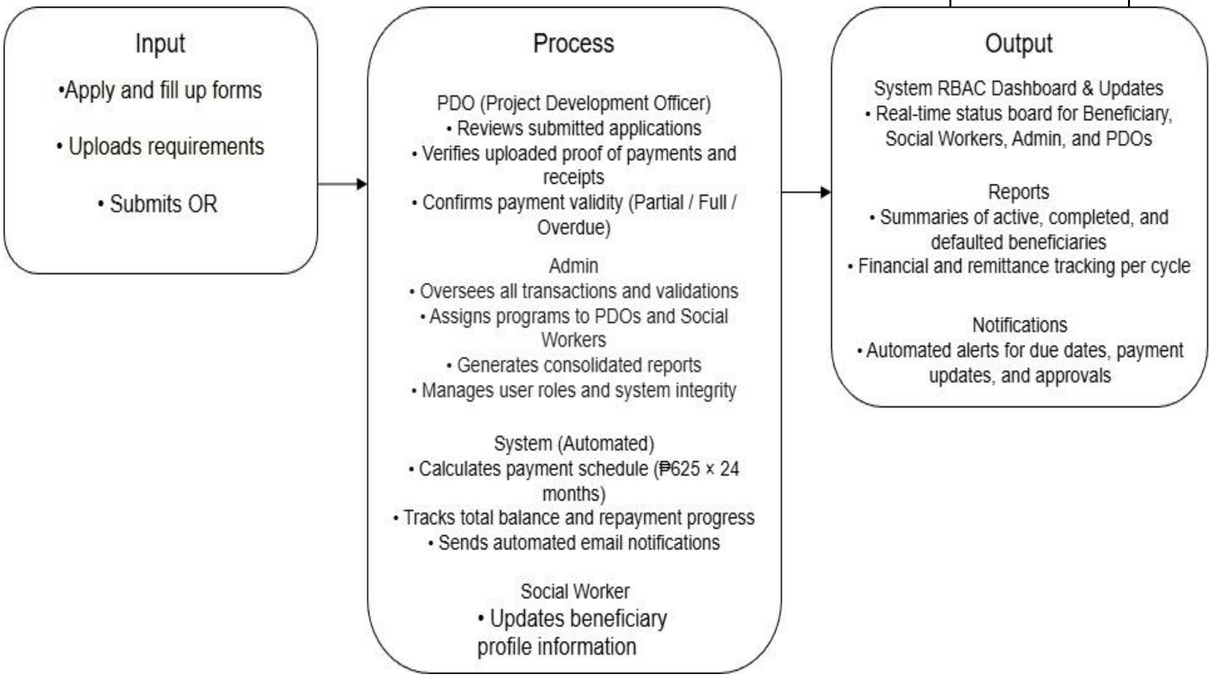

Input
•Apply and fill up forms
• Uploads requirements
• Submits OR
Process
PDO (Project Development Officer)
• Reviews submitted applications
• Verifies uploaded proof of payments and receipts
• Confirms payment validity (Partial / Full / Overdue)
Admin
• Oversees all transactions and validations
• Assigns programs to PDOs and Social Workers
• Generates consolidated reports
• Manages user roles and system integrity
System (Automated)
• Calculates payment schedule (₱625 × 24 months)
• Tracks total balance and repayment progress
• Sends automated email notifications
Social Worker
• Updates beneficiary profile information
Output
System RBAC Dashboard & Updates
• Real-time status board for Beneficiary, Social Workers, Admin, and PDOs
Reports
• Summaries of active, completed, and defaulted beneficiaries
• Financial and remittance tracking per cycle
Notifications
• Automated alerts for due dates, payment updates, and approvals

*Figure 3. Conceptual Framework (IPO Diagram)*

Figure 3 presents the Input-Process-Output (IPO) diagram of the SMART LEAP system, illustrating the overall flow of information from data entry to system-generated results. The system accepts inputs such as accomplished application forms, uploaded requirements, and submitted official receipts. These inputs are processed through application review and payment verification by the Project Development Officer, transaction supervision, role assignment, and report generation by the administrator, automated payment scheduling, balance tracking, and email notifications by the system, and beneficiary profile updates by social workers. As outputs, the system provides a role-based access control dashboard with real-time status updates, consolidated reports on beneficiary and financial records, and automated notifications for due dates, payment updates, and approvals. This conceptual diagram clearly defines the system boundaries, user roles, and data flow, ensuring transparency, accountability, and efficient system operation.

## 2.5 Use Case Diagram

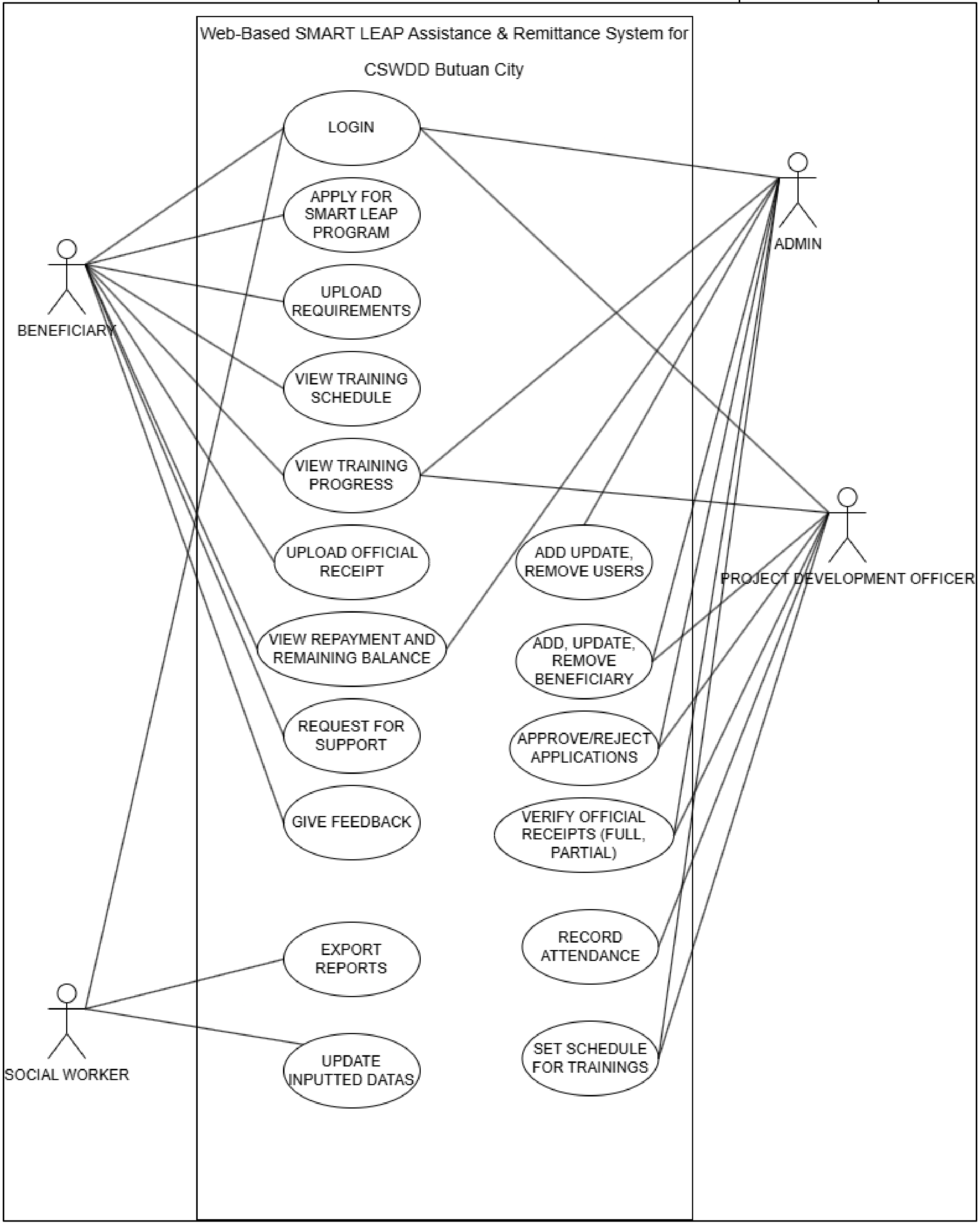

Web-Based SMART LEAP Assistance & Remittance System for CSWDD Butuan City
LOGIN
APPLY FOR SMART LEAP PROGRAM
UPLOAD REQUIREMENTS
VIEW TRAINING SCHEDULE
VIEW TRAINING PROGRESS
UPLOAD OFFICIAL RECEIPT
VIEW REPAYMENT AND REMAINING BALANCE
REQUEST FOR SUPPORT
GIVE FEEDBACK
EXPORT REPORTS
UPDATE INPUTTED DATAS
ADD UPDATE, REMOVE USERS
ADD, UPDATE, REMOVE BENEFICIARY
APPROVE/REJECT APPLICATIONS
VERIFY OFFICIAL RECEIPTS (FULL, PARTIAL)
RECORD ATTENDANCE
SET SCHEDULE FOR TRAININGS
BENEFICIARY
SOCIAL WORKER
ADMIN
PROJECT DEVELOPMENT OFFICER

*Figure 4. Use Case Diagram*

Figure 4 depicts the Web-Based SMART LEAP Assistance & Remittance System for CSWDD Butuan City use case graphic that provides an organized representation of how stakeholders engage with the System's main features. Each of the four leading actors—the Beneficiary, the Administrator, the Project Development Officer (PDO), and the Social Worker—has specific duties corresponding to their operational roles within the program. As the primary service user, the Beneficiary uses the system to fill up and submit application forms, examine training schedules and program requirements, upload official receipts, track remittances and remaining balances, and interact with the system by sending support requests and providing feedback. These exchanges place a strong emphasis on transparency, self-service, and accountability.

Maintaining operational oversight and system integrity depends on the Administrator. They are in charge of adding, updating, and removing users, keeping an eye on beneficiary applications, and verifying official receipts, and can perform any function inside the system. This role ensures that system data is kept current and user access conforms to institutional protocols. By sharing accountability for confirming official receipts and strengthening checks and balances in cash disbursement procedures, the Project Development Officer aids in the financial validation process. The PDOs also respond to beneficiary assistance enquiries and record training session attendance, and set schedules for training. Social Workers can only update inputted data and export reports. The graphic shows a clearly defined, role-based access hierarchy that promotes user empowerment, data integrity, and operational efficiency. The approach streamlines service delivery across multiple administrative levels and encourages accountability by explicitly defining who is responsible for what.

## 2.6 Activity Diagram

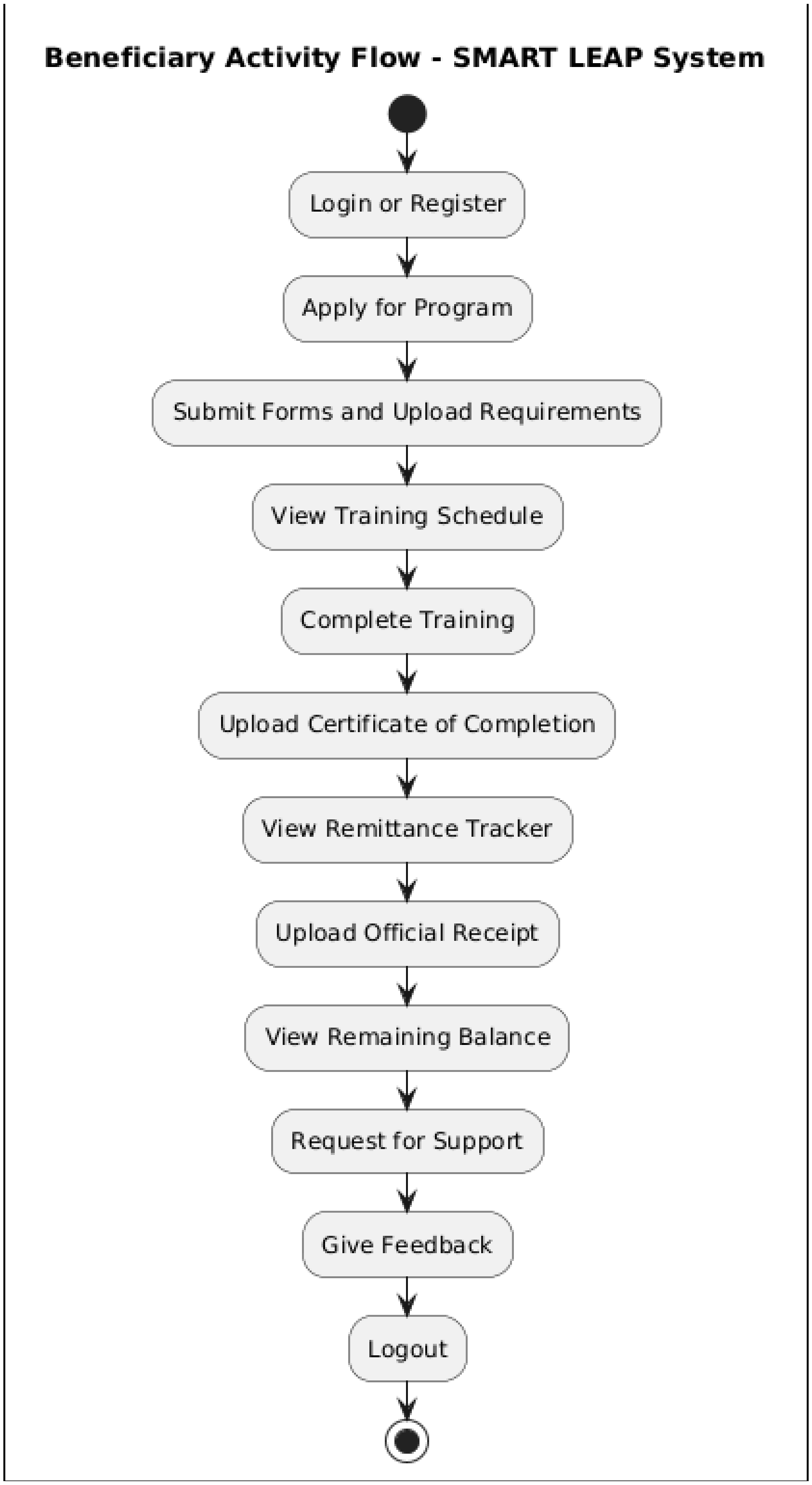

Beneficiary Activity Flow - SMART LEAP System
Login or Register
Apply for Program
Submit Forms and Upload Requirements
View Training Schedule
Complete Training
Upload Certificate of Completion
View Remittance Tracker
Upload Official Receipt
View Remaining Balance
Request for Support
Give Feedback
Logout

*Figure 5. Activity Diagram (Beneficiary)*

Figure 5 illustrates the SMART LEAP System's beneficiary activity flow, showing a clear and easy-to-follow path that guides individuals through each step of their program participation. It starts with users logging in or signing up, ensuring only authorized individuals can access the site. Once inside, recipients can enroll in a program and see what they need to do immediately. The process helps users comprehend what they need to do before moving on. Afterward, they check the training schedule and complete the necessary sessions, after which they upload their certificate of completion as proof. The remittance tracker is a part of the system that enables users to track payments and upload certified documents to hold individuals accountable.

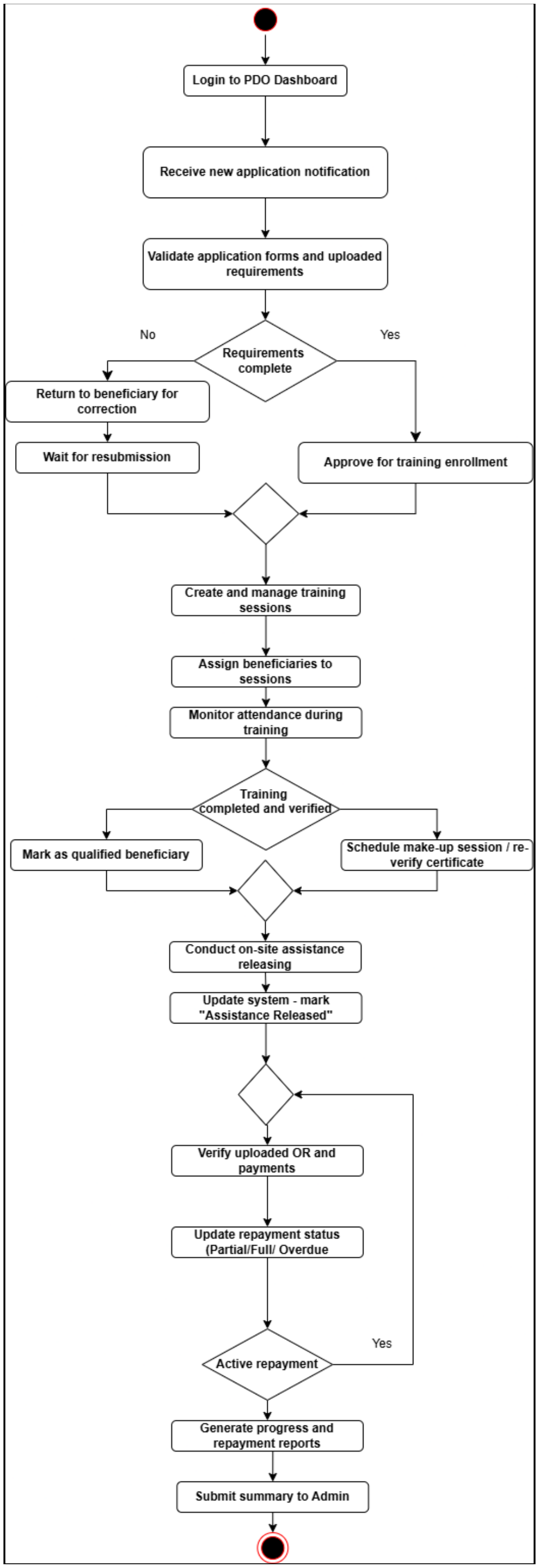

Login to PDO Dashboard
Receive new application notification
Validate application forms and uploaded requirements
No
Yes
Requirements complete
Return to beneficiary for correction
Wait for resubmission
Approve for training enrollment
Create and manage training sessions
Assign beneficiaries to sessions
Monitor attendance during training
Training completed and verified
Mark as qualified beneficiary
Schedule make-up session / re-verify certificate
Conduct on-site assistance releasing
Update system - mark "Assistance Released"
Verify uploaded OR and payments
Update repayment status (Partial/Full/ Overdue
Yes
Active repayment
Generate progress and repayment reports
Submit summary to Admin

*Figure 6 Activity Diagram* ***(****PDO)*

Figure 6 illustrates the SMART LEAP system's PDO Activity Flow, which outlines a comprehensive approach to handling beneficiary applications, training, assistance, and tracking repayments. It begins with receiving a notification about a new application and verifying that the forms and eligibility requirements are accurate. The PDO sends the application back to the Beneficiary for corrections and waits for the Beneficiary to resubmit it if the conditions are unmet. After reviewing the information, the PDO approves the Beneficiary for training enrollment, sets up training sessions, and allocates participants to them. Throughout the training, the PDO checks attendance, and after completion, verifies it to label the Beneficiary as qualified.

The next step is to assist people on-site, obtain their signatures, provide them with receipts (OR), and update the system to indicate that we have provided help. The PDO then checks to see if the payback is still active and updates the repayment status, which can be partial, complete, or overdue. The PDO prepares progress and payback reports and sends the Admin a summary of whether the Repayment is ongoing. This flow ensures that we track, check, and record every step, making the delivery of social assistance more accountable and transparent.

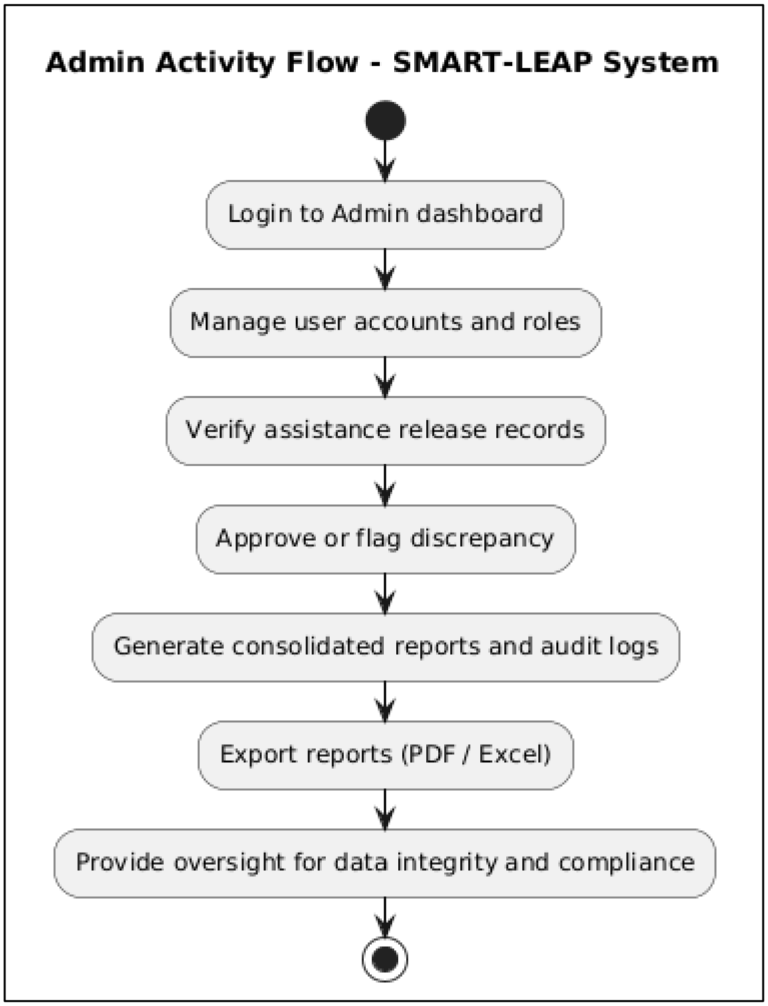

Admin Activity Flow - SMART-LEAP System
Login to Admin dashboard
Manage user accounts and roles
Verify assistance release records
Approve or flag discrepancy
Generate consolidated reports and audit logs
Export reports (PDF / Excel)
Provide oversight for data integrity and compliance

*Figure 7 Activity Diagram (Admin)*

Figure 7 illustrates the Admin Activity Flow in the SMART LEAP system, showing what system administrators need to do to keep the system running smoothly and keep an eye on it. It begins with entering the admin dashboard, managing user accounts, and assigning them the appropriate roles. Administrators then review records related to the provision of assistance to ensure that payments are accurately recorded and comply with program rules.

The Admin either approves or flags any differences for further examination if they are detected. The Admin checks the records and then creates aggregated reports and audit logs. In the final stage, the Admin ensures the data is accurate and verifies that the team follows the rules. The Admin is further responsible for maintaining an open and honest system. After providing help, the PDO updates the release summary and begins a monthly verification cycle.

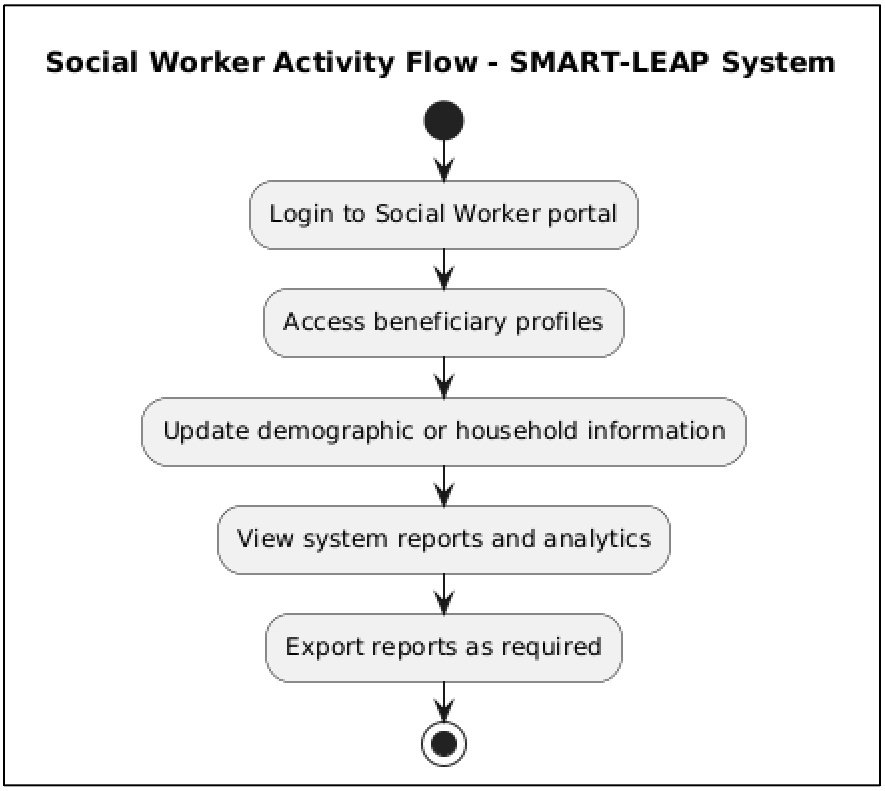

Social Worker Activity Flow - SMART-LEAP System
Login to Social Worker portal
Access beneficiary profiles
Update demographic or household information
View system reports and analytics
Export reports as required

*Figure 8 Activity Diagram (Social Worker)*

Figure 8 illustrates the SMART LEAP system's Social Worker Activity Flow, which lists the most important things social workers do to help manage beneficiary data and monitor programs.

Log in to the Social Worker portal and then proceed to each Beneficiary's profile. After that, social workers update demographic and household information to ensure that records are accurate and up-to-date. After changing the data, they look at reports and analytics that the system makes to see how well the program works and how involved the beneficiaries are. Can export these reports whenever needed for record-keeping, review, or collaboration with others interested in the project. The flow chart illustrates a streamlined, data-driven process that enhances the social worker's accountability in maintaining program integrity and ensuring services are delivered efficiently and effectively.

## 2.7 Sequence Diagram

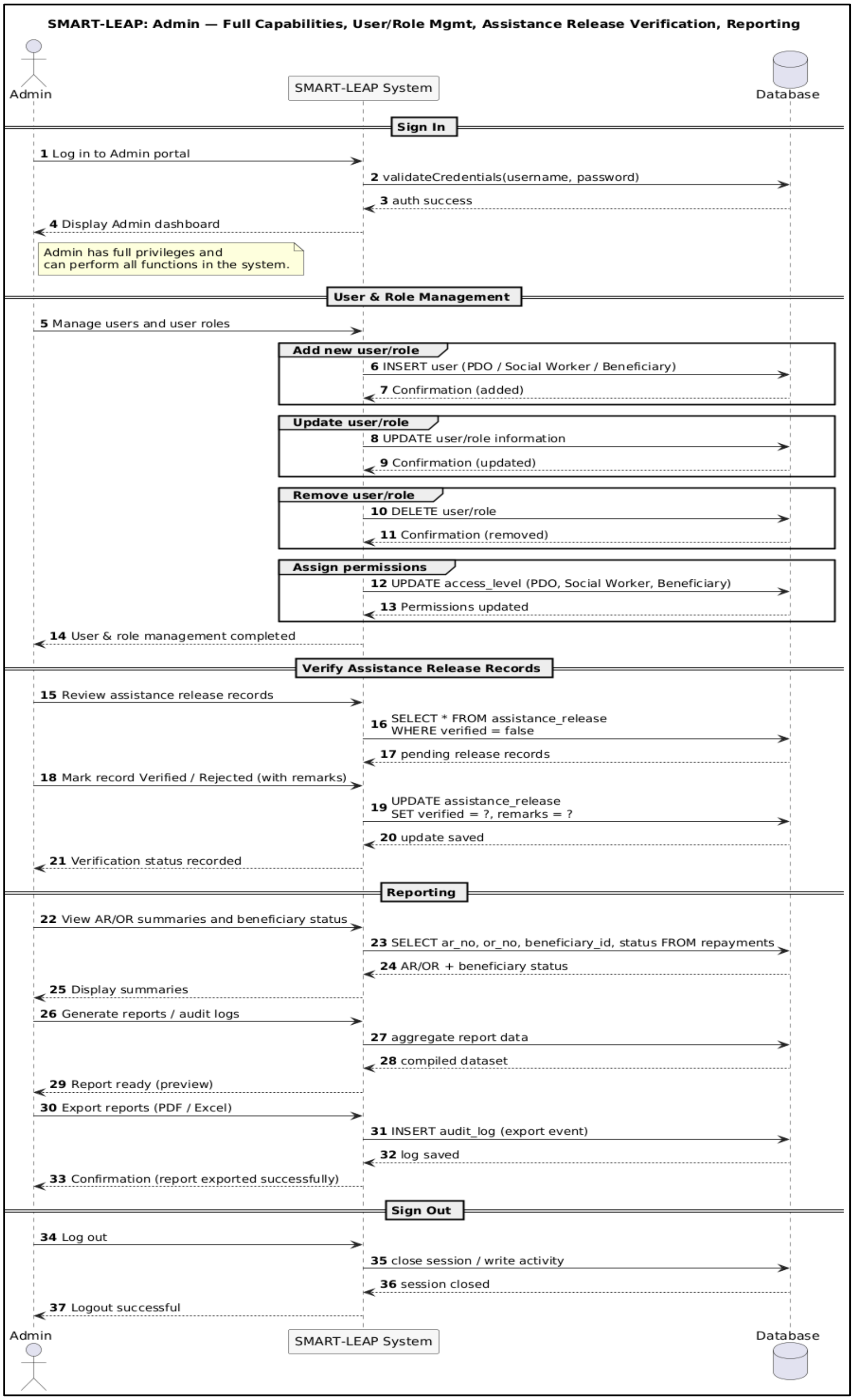

SMART-LEAP: Admin — Full Capabilities, User/Role Mgmt, Assistance Release Verification, Reporting
Admin
SMART-LEAP System
Database
Sign In
1 Log in to Admin portal
2 validateCredentials(username, password)
3 auth success
4 Display Admin dashboard
Admin has full privileges and can perform all functions in the system.
User & Role Management
5 Manage users and user roles
Add new user/role
6 INSERT user (PDO / Social Worker / Beneficiary)
7 Confirmation (added)
Update user/role
8 UPDATE user/role information
9 Confirmation (updated)
Remove user/role
10 DELETE user/role
11 Confirmation (removed)
Assign permissions
12 UPDATE access_level (PDO, Social Worker, Beneficiary)
13 Permissions updated
14 User & role management completed
Verify Assistance Release Records
15 Review assistance release records
16 SELECT * FROM assistance_release WHERE verified = false
17 pending release records
18 Mark record Verified / Rejected (with remarks)
19 UPDATE assistance_release SET verified = ?, remarks = ?
20 update saved
21 Verification status recorded
Reporting
22 View AR/OR summaries and beneficiary status
23 SELECT ar_no, or_no, beneficiary_id, status FROM repayments
24 AR/OR + beneficiary status
25 Display summaries
26 Generate reports / audit logs
27 aggregate report data
28 compiled dataset
29 Report ready (preview)
30 Export reports (PDF / Excel)
31 INSERT audit_log (export event)
32 log saved
33 Confirmation (report exported successfully)
Sign Out
34 Log out
35 close session / write activity
36 session closed
37 Logout successful
Admin
SMART-LEAP System
Database

*Figure 9 Sequence Diagram ( Admin)*

Figure 9 illustrates the Admin Sequence Diagram for the SMART-LEAP System, which depicts the interaction between the Admin, the system interface, and the Project Development Officer (PDO). The Admin's first job is to access the dashboard, where they can manage user accounts and assign tasks to ensure that everyone has the correct level of access to the platform. Once inside, the Admin sees a live summary of active users and beneficiary data, providing a quick look into how engaged people are with the system and how far the program has progressed.

The Admin can then see the most recent reports, which include descriptions of aid releases and any irregularities that staff members have noted. These reports help confirm that the finance team made payments correctly and that any unusual situations require further investigation. The Admin creates audit logs and consolidated reports to maintain transparency and accountability. Reports can be exported as PDFs or Excel files for record-keeping. These outputs serve practical purposes for internal monitoring and external reporting. Finally, the dashboard overview should confirm that it reflects all recent actions in the system.

The SMART-LEAP System also sends progress and reimbursement data to the PDO during this time, strengthening the feedback loop between administrative supervision and field-level validation. The diagram clearly indicates that the Admin is responsible for ensuring data is secure, coordinating system updates, and ensuring program operations align with compliance standards.

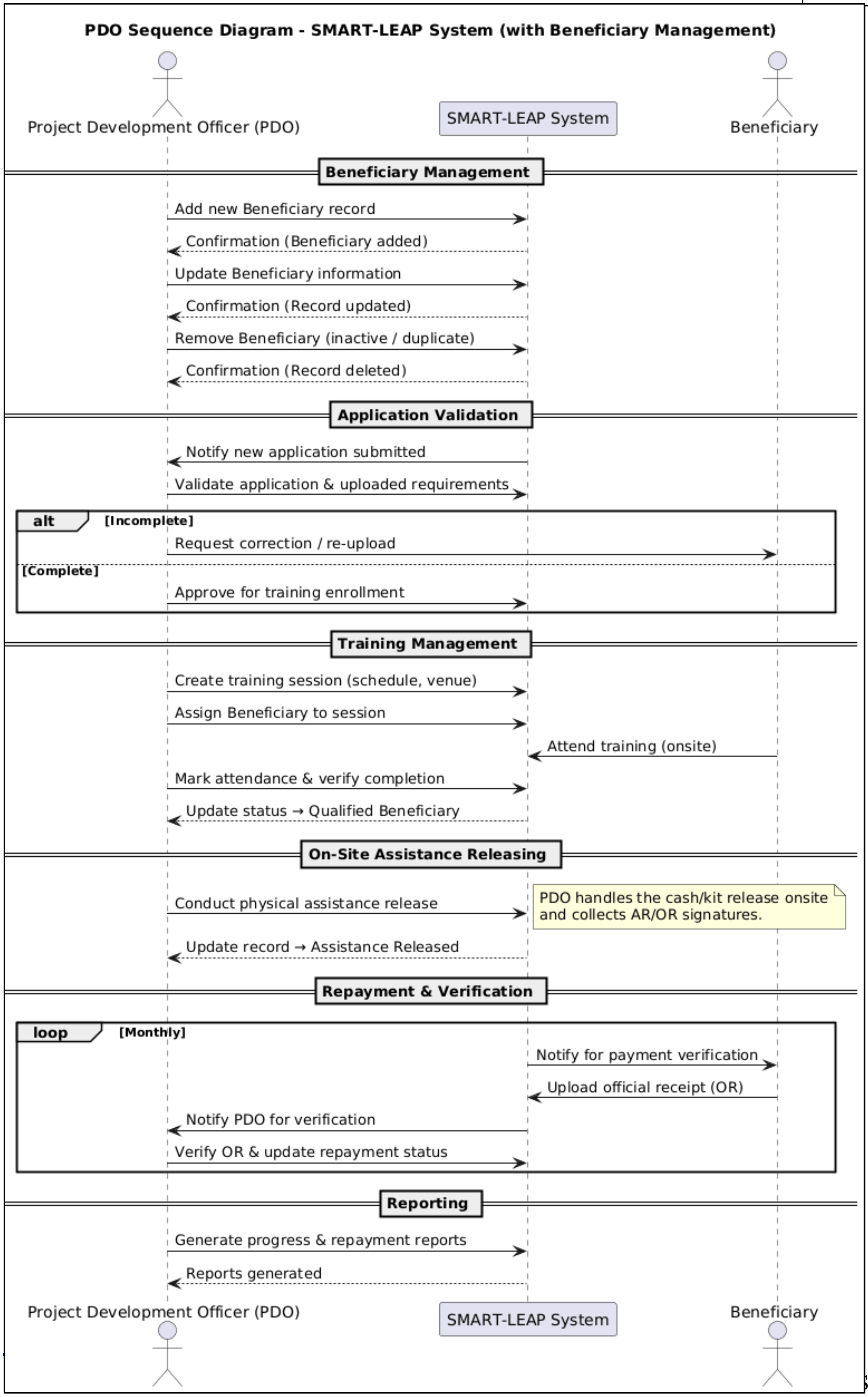

PDO Sequence Diagram - SMART-LEAP System (with Beneficiary Management)
Project Development Officer (PDO)
SMART-LEAP System
Beneficiary
Beneficiary Management
Add new Beneficiary record
Confirmation (Beneficiary added)
Update Beneficiary information
Confirmation (Record updated)
Remove Beneficiary (inactive / duplicate)
Confirmation (Record deleted)
Application Validation
Notify new application submitted
Validate application & uploaded requirements
alt
[Incomplete]
Request correction / re-upload
[Complete]
Approve for training enrollment
Training Management
Create training session (schedule, venue)
Assign Beneficiary to session
Attend training (onsite)
Mark attendance & verify completion
Update status → Qualified Beneficiary
On-Site Assistance Releasing
Conduct physical assistance release
PDO handles the cash/kit release onsite and collects AR/OR signatures.
Update record → Assistance Released
Repayment & Verification
loop
[Monthly]
Notify for payment verification
Upload official receipt (OR)
Notify PDO for verification
Verify OR & update repayment status
Reporting
Generate progress & repayment reports
Reports generated
Project Development Officer (PDO)
SMART-LEAP System
Beneficiary

*Figure 10  Sequence Diagram (PDO)*

Figure 10 illustrates the PDO Sequence Diagram for the SMART-LEAP system, outlining all the tasks assigned to the Program Development Officer, including validating applications, coordinating training, providing assistance, and managing payments. The first step is for the PDO to get a new application notification. Then, they review and approve the requirements that they submitted. After verification, the PDO sets up training sessions, assigns beneficiaries, and tracks attendance. After completing the program, we identify the beneficiaries as "Qualified." The PDO visits the site to assist with releasing and logging signatures and issuing formal receipts.

The PDO can also add, update, and remove beneficiaries from the system, in addition to these primary tasks. With this feature, the PDO can directly manage beneficiary records, ensuring that the data remains valid and up-to-date with changes in the real world, including changes in eligibility, corrections, or program exits. After providing help, the PDO updates the release summary and starts a monthly verification cycle. In this loop, beneficiaries upload receipts, and repayment trackers are updated.

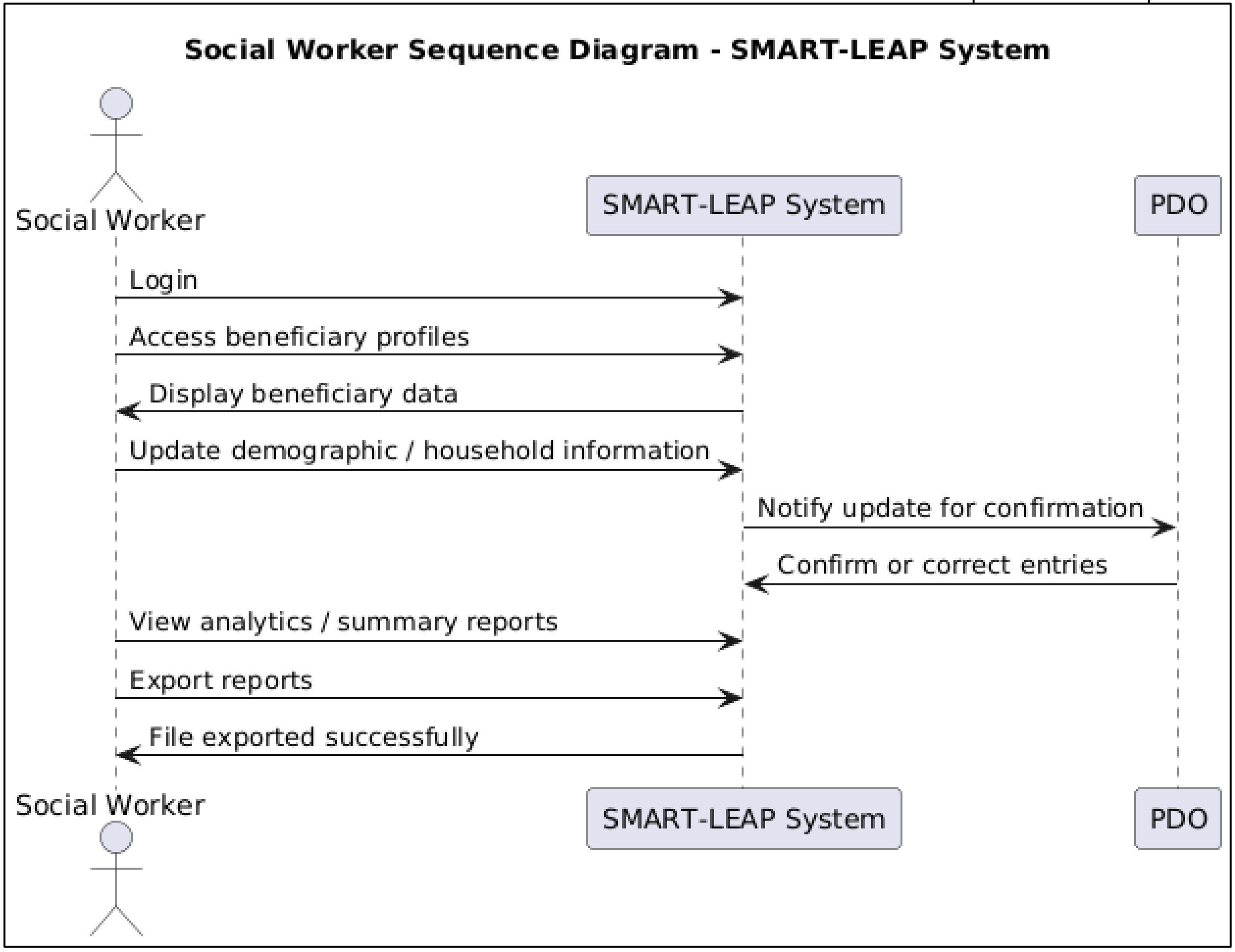

Social Worker Sequence Diagram - SMART-LEAP System
Social Worker
SMART-LEAP System
PDO
Login
Access beneficiary profiles
Display beneficiary data
Update demographic / household information
Notify update for confirmation
Confirm or correct entries
View analytics / summary reports
Export reports
File exported successfully
Social Worker
SMART-LEAP System
PDO

*Figure 11 Sequence Diagram (Social Worker)*

Figure 11 illustrates the SMART-LEAP System's Social Worker Sequence Diagram, showing how the Social Worker, the system interface, and the PDO collaborate to handle and verify data. The Social Worker starts by logging into the portal and looking at the profiles of the people they will assist. Once inside, they review and discuss the data for each recipient, focusing on demographic and household statistics.

The PDO gets an automatic message from the SMART-LEAP system after the Social Worker sends in changes. The PDO either checks the entries for accuracy or marks them for correction. We implement thorough checks and validations to ensure all beneficiary data is accurate and validated. After confirming, the Social Worker analyzes analytics and summary reports, which reveal patterns among beneficiaries, the extent of the program's reach, and the effectiveness of service delivery. The Social Worker can export these reports to maintain records or facilitate coordination. In the final stage of the process, the system verifies that it created the exported files correctly.

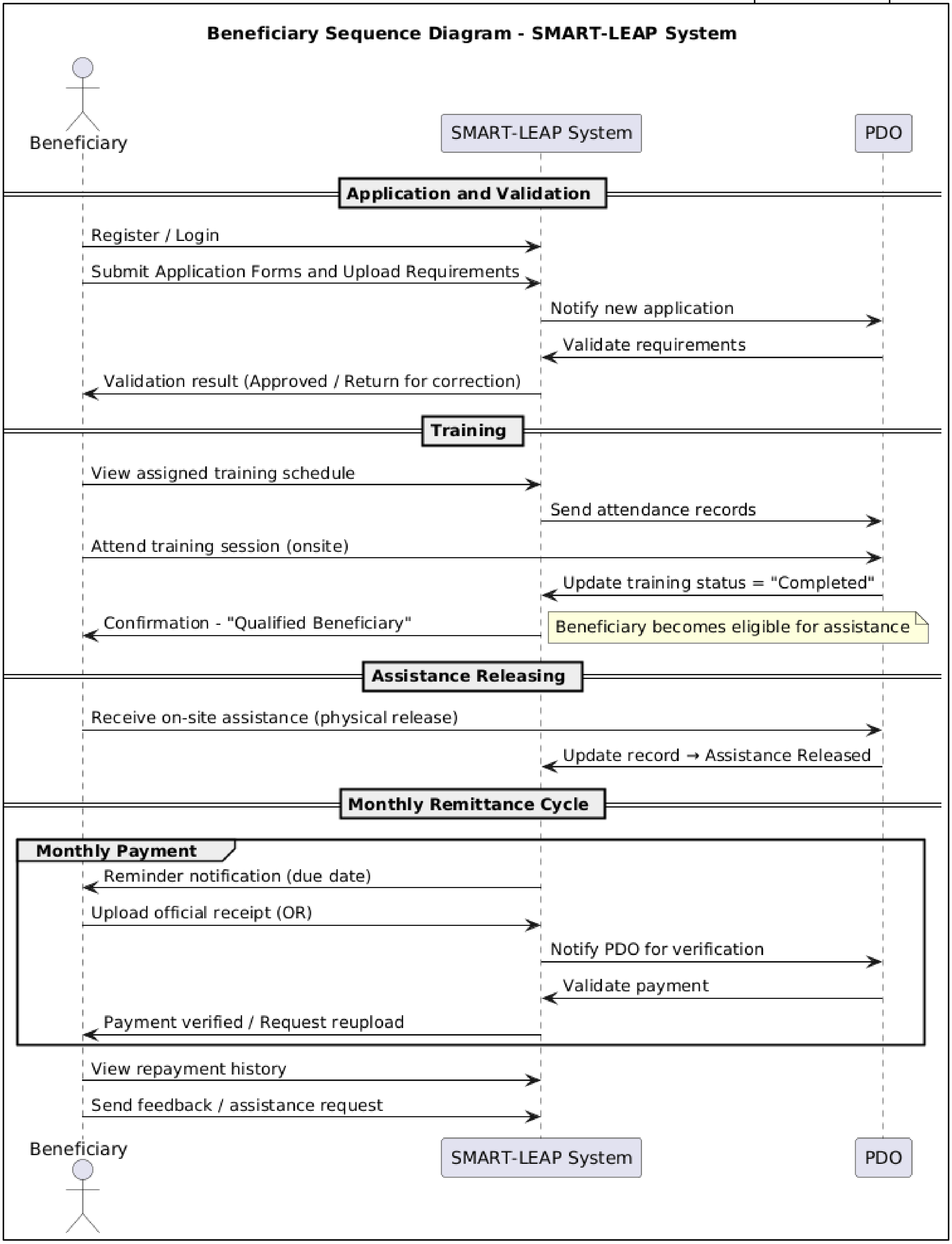

Beneficiary Sequence Diagram - SMART-LEAP System
Beneficiary
SMART-LEAP System
PDO
Application and Validation
Register / Login
Submit Application Forms and Upload Requirements
Notify new application
Validate requirements
Validation result (Approved / Return for correction)
Training
View assigned training schedule
Send attendance records
Attend training session (onsite)
Update training status = "Completed"
Confirmation - "Qualified Beneficiary"
Beneficiary becomes eligible for assistance
Assistance Releasing
Receive on-site assistance (physical release)
Update record → Assistance Released
Monthly Remittance Cycle
Monthly Payment
Reminder notification (due date)
Upload official receipt (OR)
Notify PDO for verification
Validate payment
Payment verified / Request reupload
View repayment history
Send feedback / assistance request
Beneficiary
SMART-LEAP System
PDO

***Figure 12.*** *Sequence Diagram (Beneficiary)*

Figure 12 displays the SMART-LEAP System's Beneficiary Sequence Diagram, which illustrates a beneficiary's participation process, from signing up to tracking their payments. The process starts when the person who will benefit logs into the system and fills out an application form. The SMART-LEAP system checks the submission and notifies the PDO, who then tracks attendance and updates the status to "Completed." After verifying the information, the system classifies the Beneficiary as "Qualified," meaning they are eligible for assistance. The team initiates the release of support and records it in the system along with proof of payment.

After help, the diagram changes into a monthly payment cycle. The system sends reminders to the Beneficiary to upload official receipts (OR) and check payments. These inputs update the repayment tracker, ensuring that payments are accurately tracked and recorded.

## 2.8 ERD (Database Design)

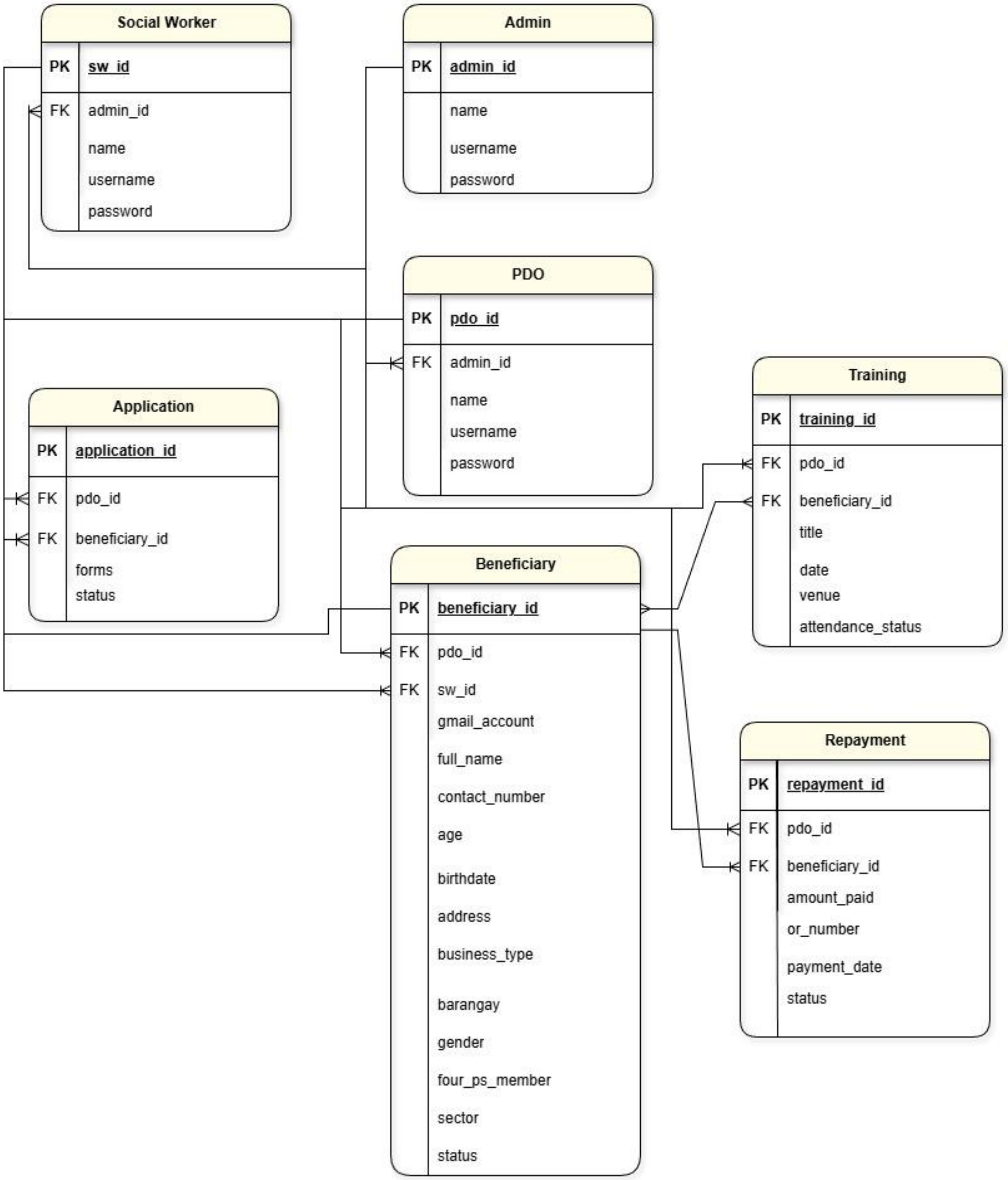

Social Worker
PK sw_id
FK admin_id
name
username
password
Admin
PK admin_id
name
username
password
PDO
PK pdo_id
FK admin_id
name
username
password
Application
PK application_id
FK pdo_id
FK beneficiary_id
forms
status
Training
PK training_id
FK pdo_id
FK beneficiary_id
title
date
venue
attendance_status
Beneficiary
PK beneficiary_id
FK pdo_id
FK sw_id
gmail_account
full_name
contact_number
age
birthdate
address
business_type
barangay
gender
four_ps_member
sector
status
Repayment
PK repayment_id
FK pdo_id
FK beneficiary_id
amount_paid
or_number
payment_date
status

*Figure 13. ERD Database Design*

Figure 13 shows the entity-relationship diagram (ERD) that illustrates the organization of the SMART LEAP database. It illustrates how the main entities—Admin, Social Worker, Project Development Officer (PDO), Beneficiary, Application, Training, and Repayment—are interconnected, for example, through applications that link Beneficiaries to PDOs or Training sessions assigned to Beneficiaries. Each entity has important characteristics and holds information needed for the system to work. By effectively managing and tracking data and monitoring program processes, we ensure that we address everything appropriately.

The admin account is the primary account that oversees the system. It stores information such as admin_id, name, username, and password. Implementing this system allows for safe authentication and unified management. Using a foreign key relationship (admin_id), the admin entity connects the social worker and PDO entities. The structure establishes a hierarchy of permissions. This arrangement guarantees that management control and system access are managed appropriately through a unified set of accounts.

The Beneficiary entity stores thorough information about each program recipient's background and finances. Full name, phone number, age, date of birth, barangay, address, business type, household size, gender, four_ps_member, sector, and position are some of the most important details. The Beneficiary entity is linked directly to the Application, Training, and Repayment modules. It is where most system tasks begin and end.

The application organization tracks beneficiaries' requests for assistance with earning a living. Each application entry—identified by application_id and beneficiary_id—records who approved by pd_id, forms, and status. The training entity links to the pdo_id, which connects the training_id, title, date, location, and attendance_status. This connection allows PDOs to track and cross-reference which beneficiaries attend training, tying attendance records to individual applications and demonstrating readiness and eagerness to participate.

The Repayment entity records payments to beneficiaries by storing the following information: repayment_id, beneficiary_id, verified_by, amount_paid, payment_date, and payment status. The Repayment entity checks all payment events and maintains records, working with the Beneficiary entity and the PDO/Admin. These links clearly illustrate how money flows between the Repayment, the Beneficiary, and the PDO/Admin, ensuring that payment records are accurate and easy to understand.

**2.9 User Interface Design (Prototype)**

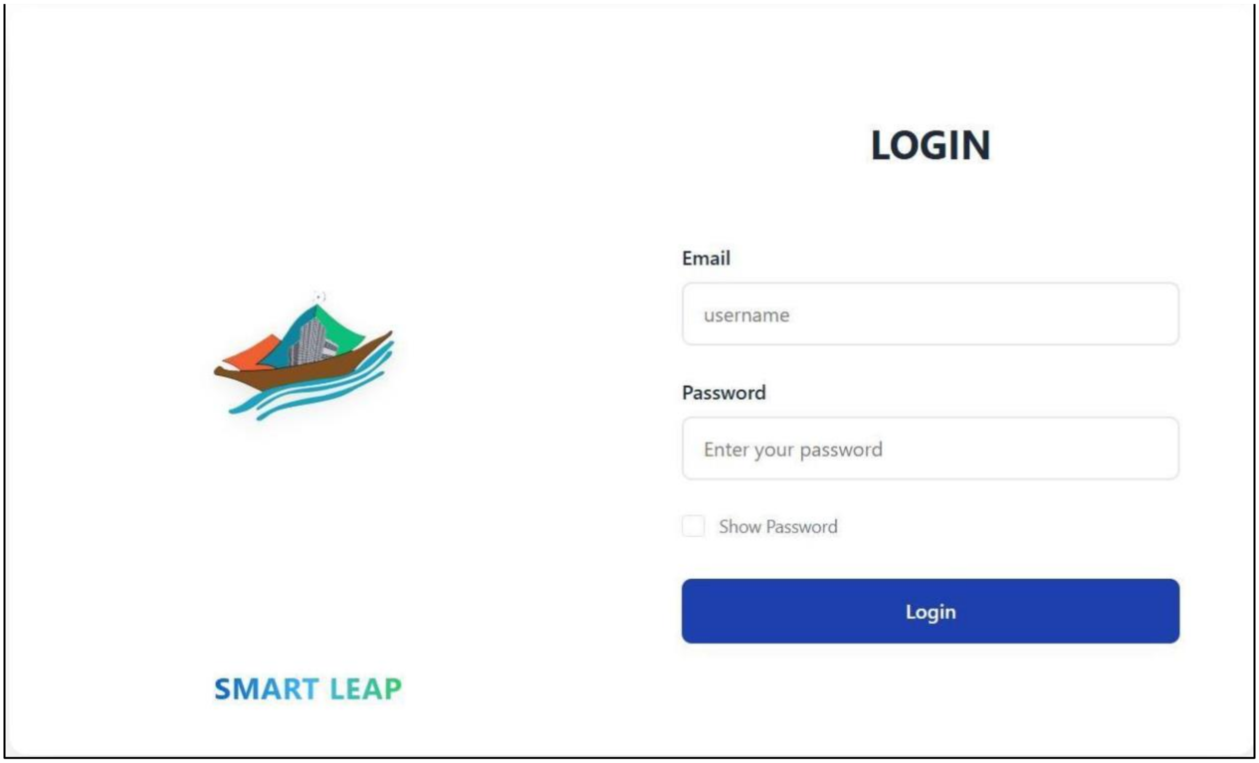


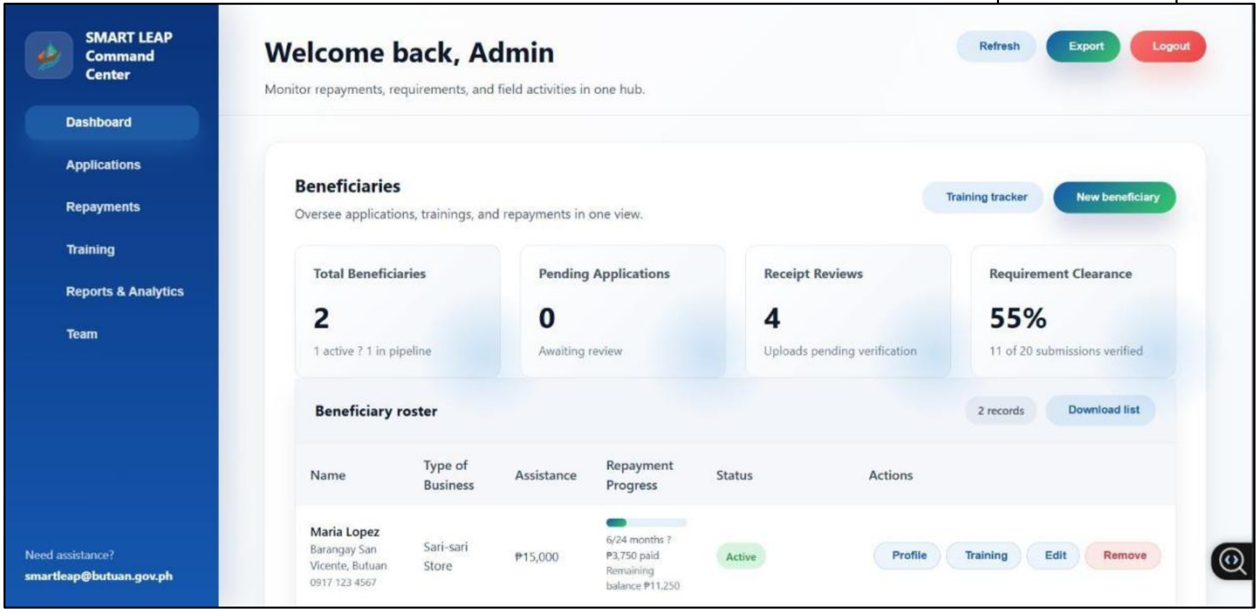
*Figure 14. Login Page (Admin, PDO, Social Worker)*

Figure 14 illustrates the login page as the entry point to the administrative side of SMART LEAP, where authorized users manage the system's operational backbone. Its job is not just to verify users; it is also a security checkpoint that ensures only verified users can access sensitive information, including beneficiary applications, financial disbursements, training records, and audit logs. The system maintains role-based access control by demanding credentials. Maintaining data security and accountability across all government operations is crucial.

The site is identical for all three roles: Admin, PDO, and Social Worker. However, once logged in, their permissions and dashboard displays differ. Admins can perform various tasks within the system, including managing users and generating reports. While PDOs are responsible for verifying workflows and overseeing beneficiaries, Social Workers are responsible for updating data and exporting reports. The unified login page makes it easier to get in, but the backend logic still filters functionality based on user type.

The design of the website simplifies the login process and reduces cognitive burden. The "Show Password" option is a nice touch that makes the program more user-friendly, especially for senior users who may struggle with concealed inputs. However, not having cues or visual differences for each position could cause confusion or misclicks. Minor adjustments could address this issue, such as implementing a role selector or adding dynamic welcome prompts.

This login page is an important part of SMART LEAP's digital infrastructure. It strikes a good mix between security and ease of use. It does its main job well, but it could be improved to reflect the unique journeys of its users better and build trust by providing more precise visual and contextual feedback.


SMART LEAP Command Center
Dashboard
Applications
Repayments
Training
Reports & Analytics
Team
Refresh
Export
Logout
Training tracker
New beneficiary
2 records
Download list
Profile
Training
Edit
Remove

Need assistance?
smartleap@butuan.gov.ph

**Welcome back, Admin**

Monitor repayments, requirements, and field activities in one hub.

**Beneficiaries**

Oversee applications, trainings, and repayments in one view.

| Total Beneficiaries | Pending Applications | Receipt Reviews | Requirement Clearance |
|---|---|---|---|
| 2 | 0 | 4 | 55% |
| 1 active ? 1 in pipeline | Awaiting review | Uploads pending verification | 11 of 20 submissions verified |

**Beneficiary roster**

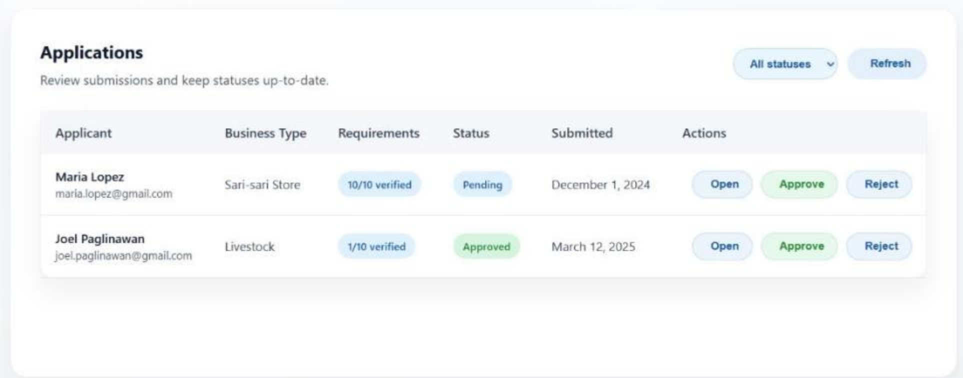
| Name | Type of Business | Assistance | Repayment Progress | Status | Actions |
|---|---|---|---|---|---|
| Maria Lopez<br>Barangay San Vicente, Butuan<br>0917 123 4567 | Sari-sari Store | ₱15,000 | 6/24 months ?<br>₱3,750 paid<br>Remaining balance ₱11,250 | Active | Profile Training Edit Remove |

***Figure 15**. Admin Dashboard*

Figure 15 illustrates the login page as the entry point to the administrative side of SMART LEAP, where authorized users manage the system's operational backbone. Its job is not just to verify users; it is also a security checkpoint that ensures only verified users can access sensitive information, including beneficiary applications, financial disbursements, training records, and audit logs. The system maintains role-based access control by demanding credentials. This is crucial for maintaining data security and accountability across all government operations.

This is the identical site for all three roles: Admin, PDO, and Social Worker. However, once they are logged in, their permissions and dashboard displays are different. Admins can perform various tasks within the system, including managing users and generating reports. While PDOs are responsible for verifying workflows and overseeing beneficiaries, Social Workers are responsible for updating data and exporting reports. The unified login page makes it easier to get in, but the backend logic still filters functionality based on user type.

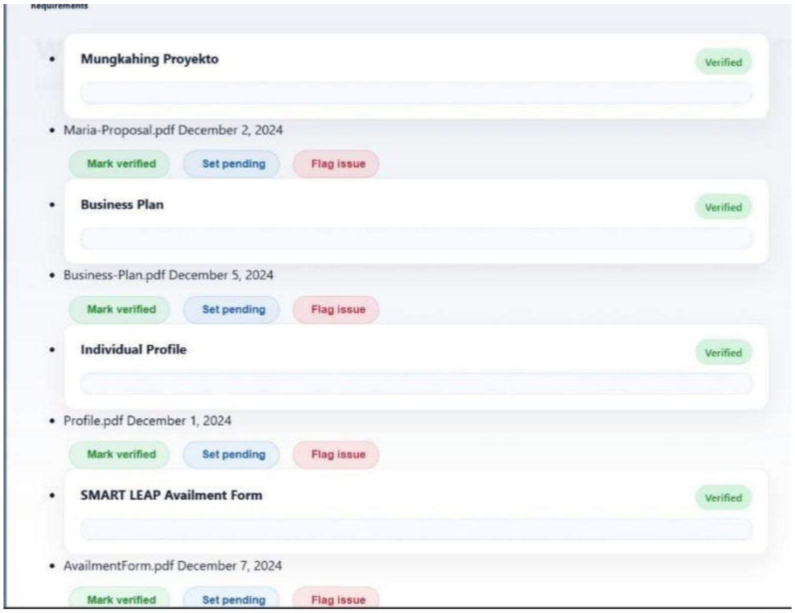
The website is intentionally designed to be simple, facilitating easier login and reducing cognitive burden. The "Show Password" option is a nice touch that makes the program more user-friendly, especially for senior users who may struggle with concealed inputs. However, not having cues or visual differences for each position could cause confusion or misclicks. This could be addressed with minor adjustments, such as a role selector or dynamic welcome prompts. This login page is an important part of SMART LEAP's digital infrastructure. It strikes a good mix between security and ease of use. It does its main job well, but it could be improved to reflect the unique journeys of its users better and build trust by providing more precise visual and contextual feedback.



**Applications**

Review submissions and keep statuses up-to-date.

| Applicant | Business Type | Requirements | Status | Submitted | Actions |
|---|---|---|---|---|---|
| **Maria Lopez**<br>maria.lopez@gmail.com | Sari-sari Store | 10/10 verified | Pending | December 1, 2024 | Open Approve Reject |
| **Joel Paglinawan**<br>joel.paglinawan@gmail.com | Livestock | 1/10 verified | Approved | March 12, 2025 | Open Approve Reject |



Requirements

- **Mungkahing Proyekto** Verified
- Maria-Proposal.pdf December 2, 2024
- **Business Plan** Verified
- Business-Plan.pdf December 5, 2024
- **Individual Profile** Verified
- Profile.pdf December 1, 2024
- **SMART LEAP Availment Form** Verified
- AvailmentForm.pdf December 7, 2024

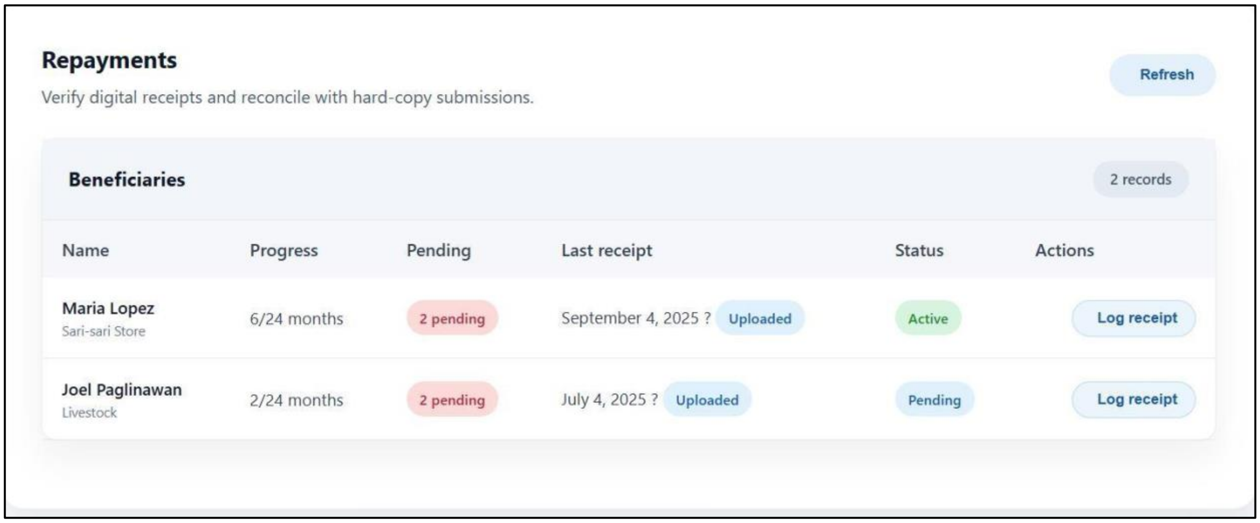
*Figure 16. Admin & PDO Application Management*

Figure 16 shows that this dashboard view gives a quick overview of all the applications filed. Admins can view who has applied, the type of business they wish to start, the level of detail in their requirements, and the current status of their application. The tabular style makes it easy to scan and compare, while options like "Open" and "Report" allow you to examine details more closely or take action. This interface helps admins triage and monitor apps by determining which ones need attention, which are ready for approval, and which may require follow-up.

This dashboard provides a quick overview of several beneficiaries, including their repayment progress, pending receipts, last receipt dates, and current status. It was designed for quick screening and triage, allowing Admins or PDOs to see who is on track, who has overdue receipts, and who requires follow-up. The "Log receipt" action button lets you make changes immediately, making this interface ideal for field verification or batch processing.

This view focuses on one Beneficiary's financial path, providing extensive information about the amount of help they received, the amount they have repaid, the amount they still owe, and their current balance. The payment history table is quite helpful. It displays transactions by month, the status of receipts, and actions such as "Approve," "Needs review," or "Remove."  Admins can use this interface to manage cases and prepare for audits. They can verify receipts, ensure that digital and hard-copy submissions match, and confirm that program requirements are satisfied.

The view provides detailed information about the materials that each applicant submitted, displaying the file names, submission dates, and status indicators such as "Verified," "Set pending," "Mark verified," and "Flag issue." Admins or PDOs conduct detailed validation by verifying that each necessary file meets the program's specifications.



**Repayments**

Verify digital receipts and reconcile with hard-copy submissions.

**Beneficiaries**

2 records

| Name | Progress | Pending | Last receipt | Status | Actions |
|---|---|---|---|---|---|
| Maria Lopez<br>Sari-sari Store | 6/24 months | 2 pending | September 4, 2025 ? Uploaded | Active | Log receipt |
| Joel Paglinawan<br>Livestock | 2/24 months | 2 pending | July 4, 2025 ? Uploaded | Pending | Log receipt |

*Figure 17. Repayments (Admin & PDO)*

Figure 17 illustrates that this dashboard provides a quick overview of several beneficiaries, including their repayment progress, pending receipts, last receipt dates, and current status. It was designed for quick screening and triage, allowing Admins or PDOs to see who is on track, who has overdue receipts, and who requires follow-up. The "Log receipt" action button lets you make changes immediately, making this interface ideal for field verification or batch processing.

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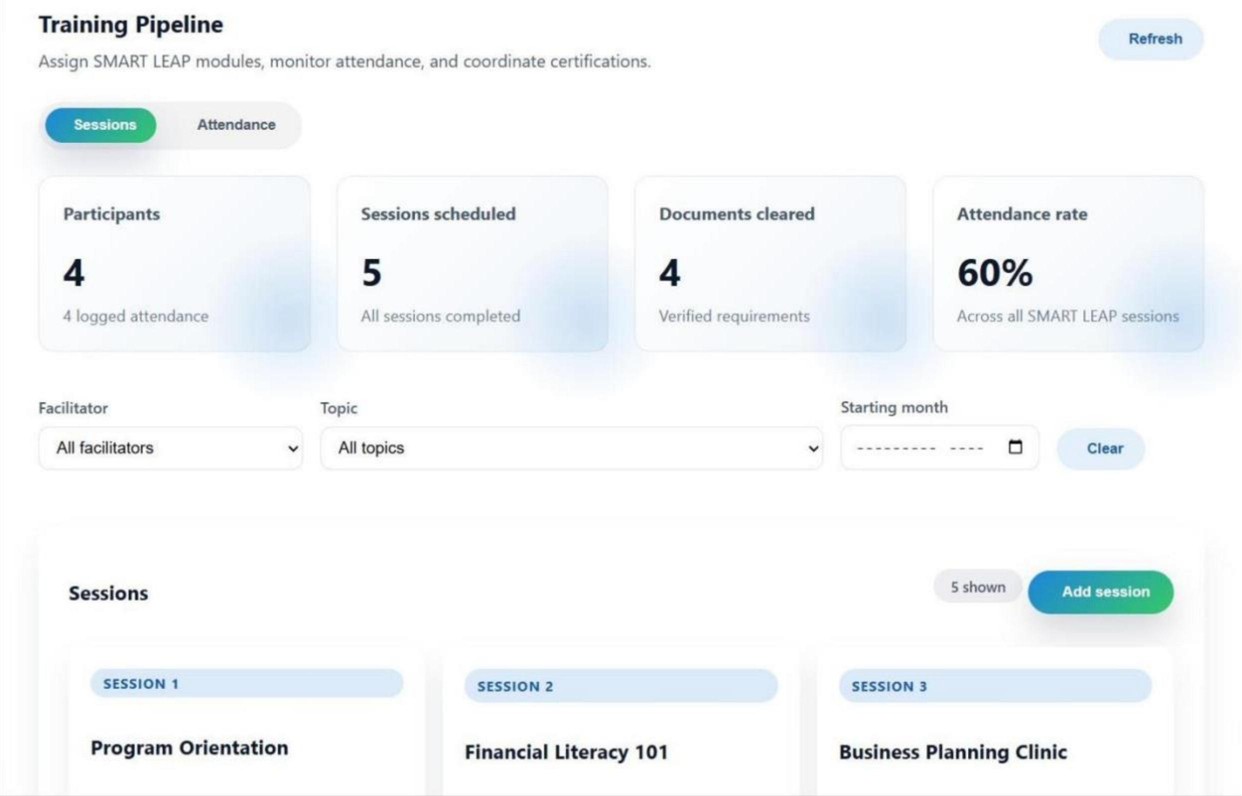


**Training Pipeline**

Assign SMART LEAP modules, monitor attendance, and coordinate certifications.

| Participants | Sessions scheduled | Documents cleared | Attendance rate |
|---|---|---|---|
| 4 | 5 | 4 | 60% |
| 4 logged attendance | All sessions completed | Verified requirements | Across all SMART LEAP sessions |

Facilitator: All facilitators
Topic: All topics
Starting month: --------- ----

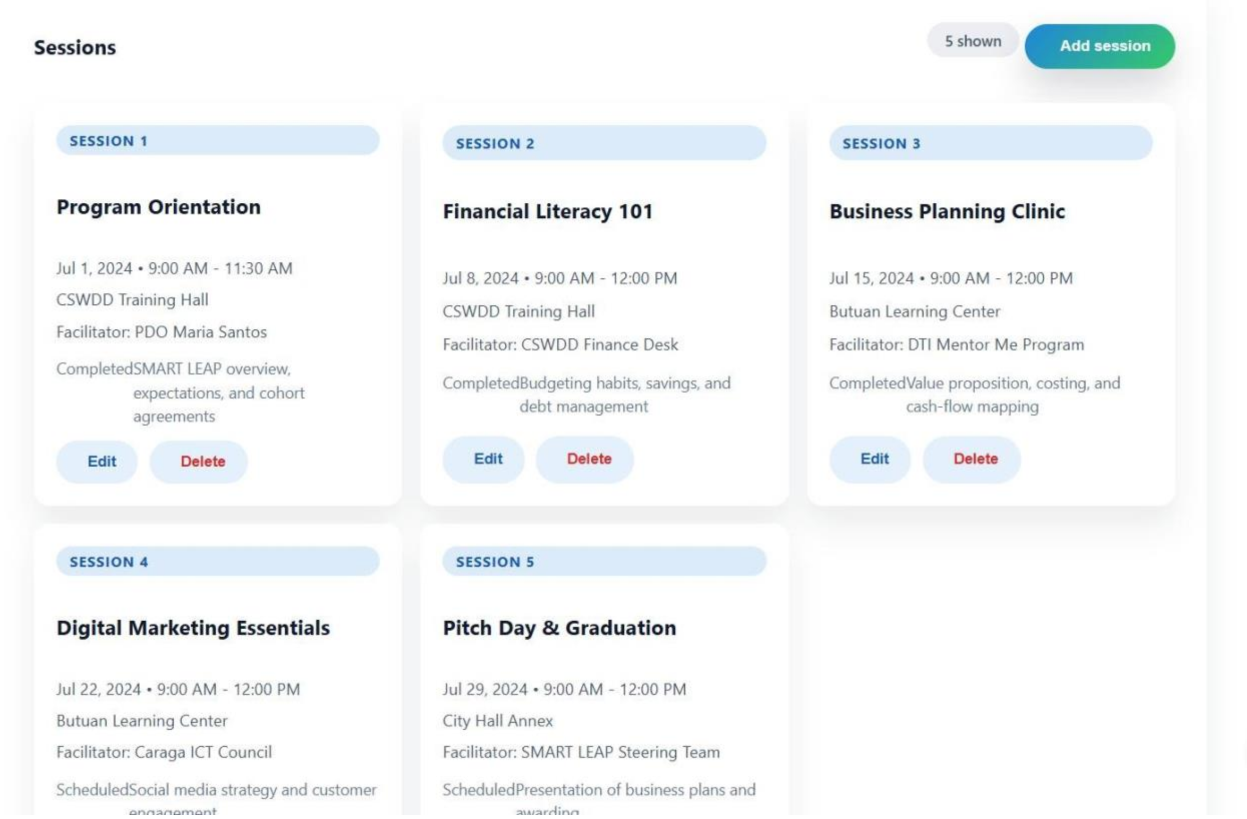
**Sessions**

5 shown

SESSION 1
**Program Orientation**

SESSION 2
**Financial Literacy 101**

SESSION 3
**Business Planning Clinic**



**Sessions**

5 shown

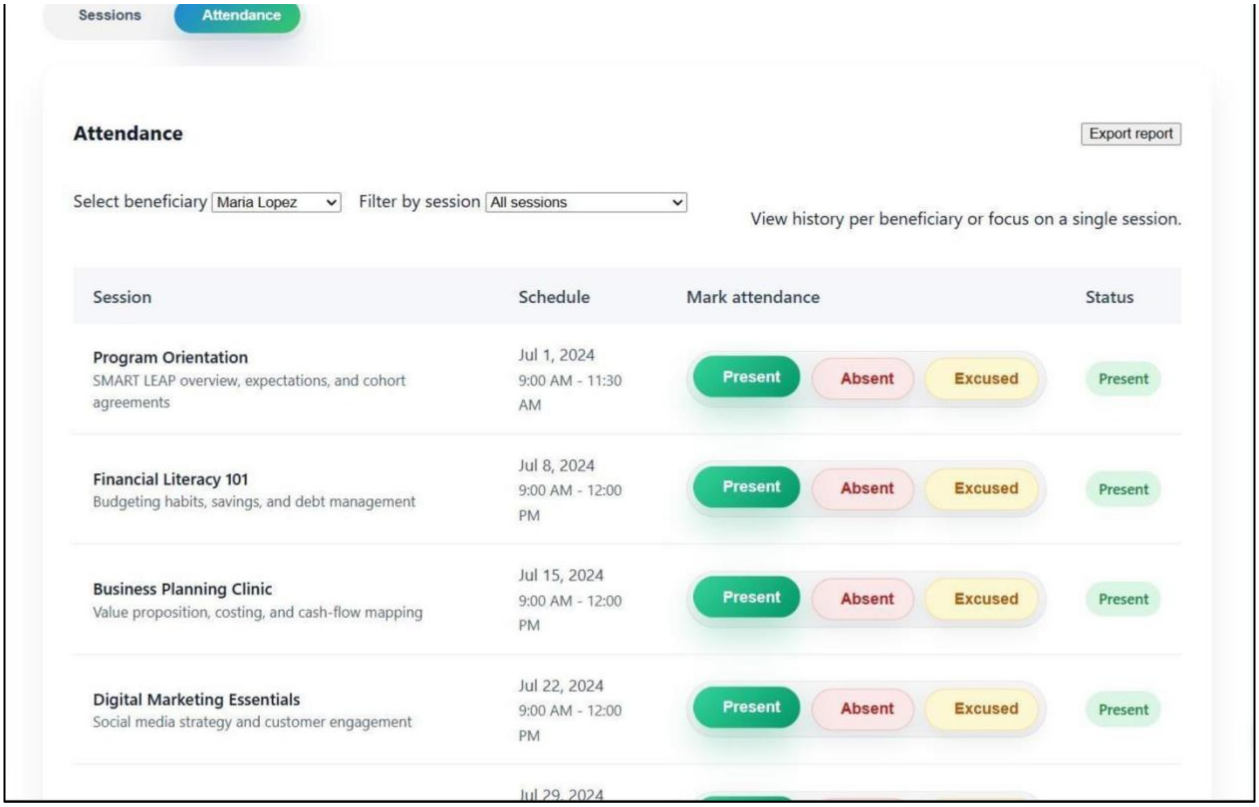
SESSION 1
**Program Orientation**
Jul 1, 2024 • 9:00 AM - 11:30 AM
CSWDD Training Hall
Facilitator: PDO Maria Santos
CompletedSMART LEAP overview, expectations, and cohort agreements

SESSION 2
**Financial Literacy 101**
Jul 8, 2024 • 9:00 AM - 12:00 PM
CSWDD Training Hall
Facilitator: CSWDD Finance Desk
CompletedBudgeting habits, savings, and debt management

SESSION 3
**Business Planning Clinic**
Jul 15, 2024 • 9:00 AM - 12:00 PM
Butuan Learning Center
Facilitator: DTI Mentor Me Program
CompletedValue proposition, costing, and cash-flow mapping

SESSION 4
**Digital Marketing Essentials**
Jul 22, 2024 • 9:00 AM - 12:00 PM
Butuan Learning Center
Facilitator: Caraga ICT Council
ScheduledSocial media strategy and customer engagement

SESSION 5
**Pitch Day & Graduation**
Jul 29, 2024 • 9:00 AM - 12:00 PM
City Hall Annex
Facilitator: SMART LEAP Steering Team
ScheduledPresentation of business plans and awarding



**Attendance**

Select beneficiary: Maria Lopez
Filter by session: All sessions

View history per beneficiary or focus on a single session.

| Session | Schedule | Mark attendance | Status |
|---|---|---|---|
| **Program Orientation**<br>SMART LEAP overview, expectations, and cohort agreements | Jul 1, 2024<br>9:00 AM - 11:30 AM | Present Absent Excused | Present |
| **Financial Literacy 101**<br>Budgeting habits, savings, and debt management | Jul 8, 2024<br>9:00 AM - 12:00 PM | Present Absent Excused | Present |
| **Business Planning Clinic**<br>Value proposition, costing, and cash-flow mapping | Jul 15, 2024<br>9:00 AM - 12:00 PM | Present Absent Excused | Present |
| **Digital Marketing Essentials**<br>Social media strategy and customer engagement | Jul 22, 2024<br>9:00 AM - 12:00 PM | Present Absent Excused | Present |
|  | Jul 29, 2024 |  |  |

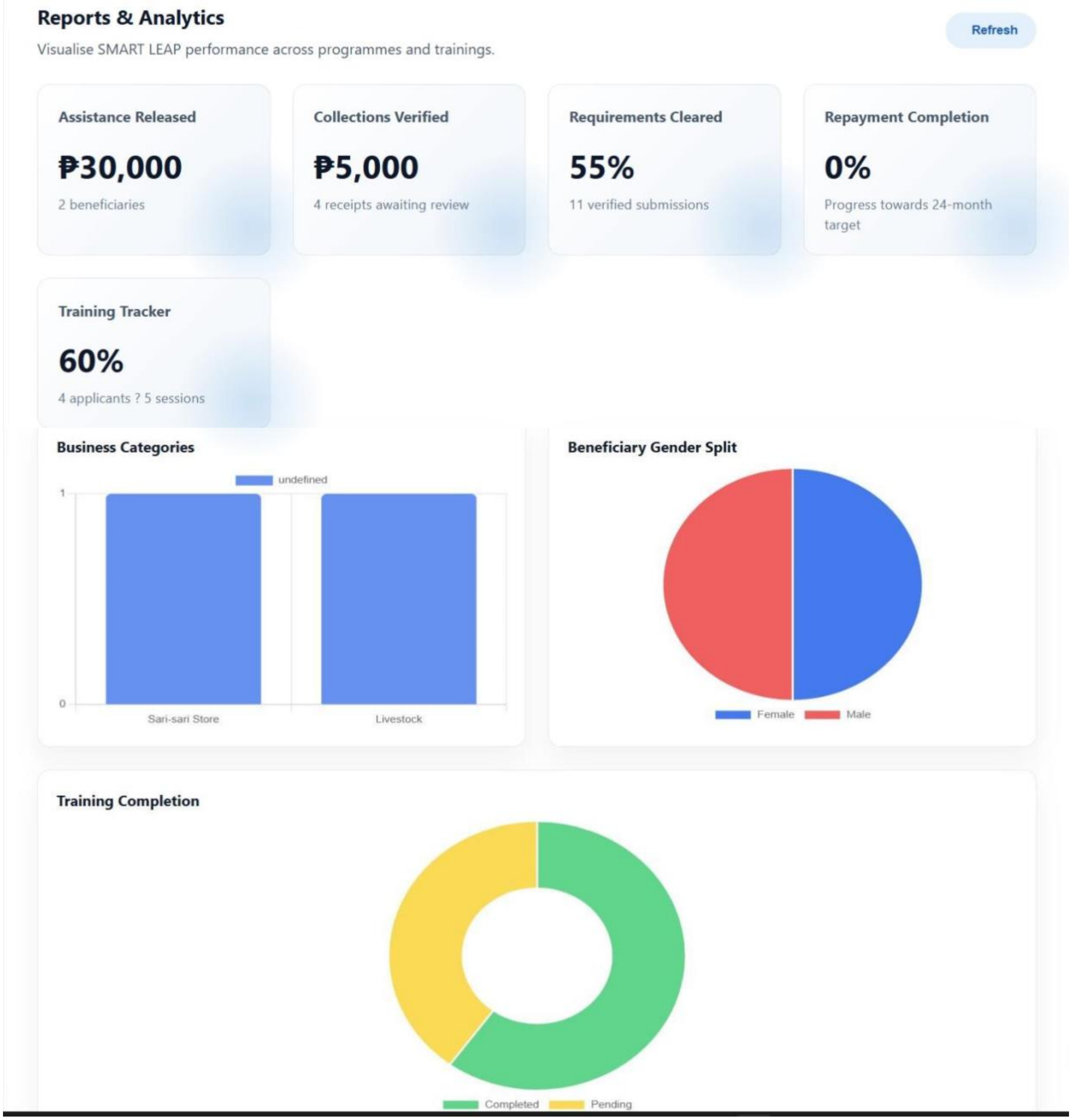
*Figure 18. Training (Admin & PDO)*

Figure 18 illustrates that this interface enables admin and PDOs to view the status of all sessions and participants in their training at a glance. It displays important statistics, including the number of completed sessions, the attendance rate, and the number of documents processed. Admin and PDOs can use the dropdown filters to break down data by facilitator, topic, or time period. The dashboard simplifies the process of identifying gaps and trends in participation. The program is structured to oversee activities at the program level, enabling decision-makers to evaluate whether the training team achieves the goals and to identify areas that require further

attention.

This view focuses on the logistics and content of each training session. Each session can be viewed as a card, displaying information such as the title, date, venue, facilitator, and description. Using the "Add session" option, administrators can modify, delete, or add new sessions. This interface facilitates planning and organizing sessions, ensuring they are well-documented, staffed correctly, and aligned with SMART LEAP's learning goals.

This interface allows PDOs to designate attendance for each session as "Present," "Absent," or "Excused." It zooms in on each person's participation. Beneficiaries and sessions can filter it, and it includes an export option for reporting purposes. This tool is essential for ensuring that people follow the rules, especially when completing training is required for receiving financial assistance or certification.



**Reports & Analytics**

Visualise SMART LEAP performance across programmes and trainings.

| | | | |
|---|---|---|---|
| Assistance Released | Collections Verified | Requirements Cleared | Repayment Completion |
| ₱30,000 | ₱5,000 | 55% | 0% |
| 2 beneficiaries | 4 receipts awaiting review | 11 verified submissions | Progress towards 24-month target |

Training Tracker
60%
4 applicants ? 5 sessions

Refresh
Business Categories
undefined
1
0
Sari-sari Store
Livestock
Beneficiary Gender Split
Female
Male
Training Completion
Completed
Pending

*Figure 19. Reports & Analytics (Admin)*

Figure 17 illustrates that this dashboard lets admin view the SMART LEAP program's most critical performance indicators in real time. It combines parameters such as help provided, verified collections, cleared requirements, progress on payback, and completed training into easy-to-read data blocks. Each metric shows a different aspect of the program's health:

- Financial Flow: The amount of help that organizations provide and the amount that recipients receive.

- Compliance: How many of the team members checked the requirements?

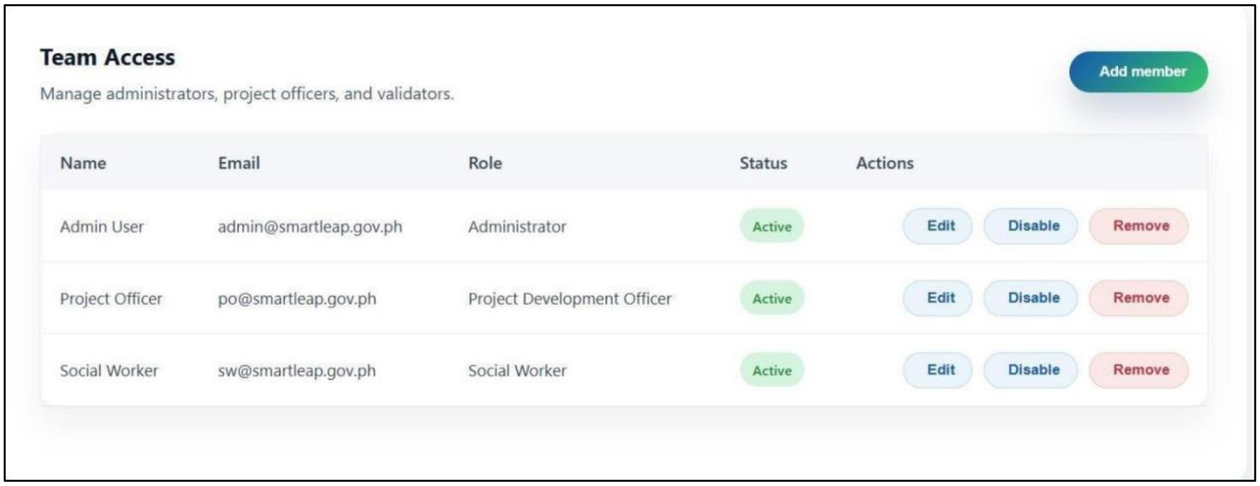
- Engagement: How far along are the recipients in completing their training and making payments?

This interface integrates with the dashboard by converting data into visual representations, such as bar charts, pie charts, and donut charts. These pictures make it simpler to see patterns and differences:

- Business Categories: Shows the kinds of businesses that can get help (for example, sari-sari stores and livestock).

- Gender Split: How well are the beneficiaries balanced in demographics?

- Completion of Training: Displays progress visually across sessions that we have completed, are still pending, or have not started yet.



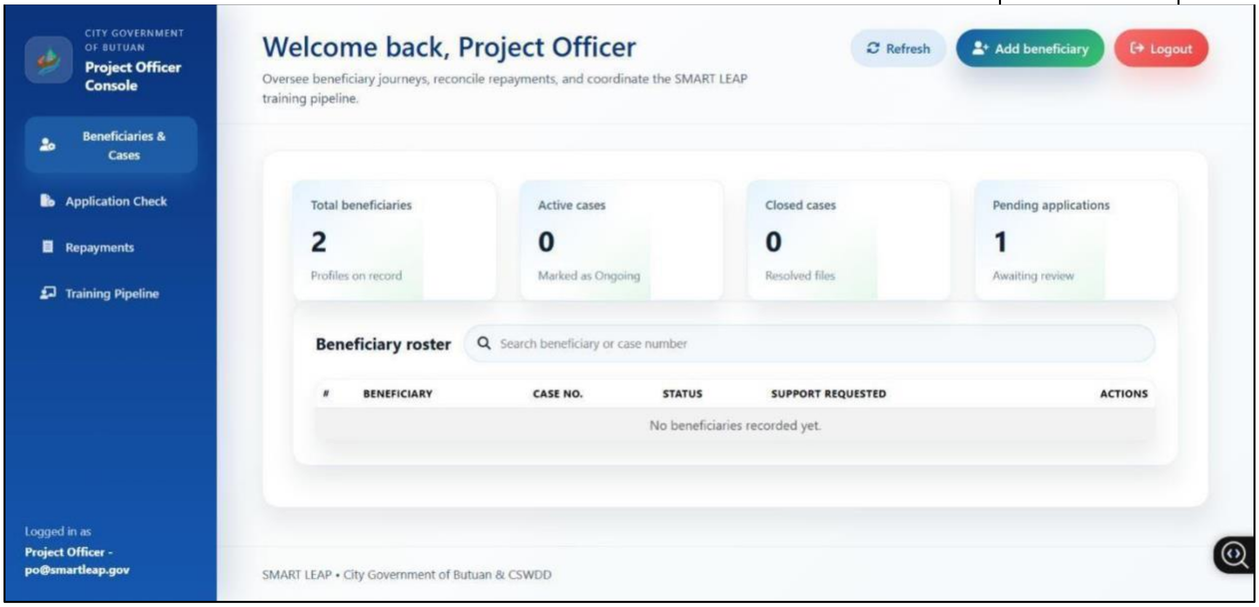
**Team Access**

Manage administrators, project officers, and validators.

| Name | Email | Role | Status | Actions |
|---|---|---|---|---|
| Admin User | admin@smartleap.gov.ph | Administrator | Active | Edit Disable Remove |
| Project Officer | po@smartleap.gov.ph | Project Development Officer | Active | Edit Disable Remove |
| Social Worker | sw@smartleap.gov.ph | Social Worker | Active | Edit Disable Remove |

*Figure 20. User Management (Admin)*

Figure 20 shows that administrators can manage user accounts and set access levels for different roles in the system using the Team Access section of the SMART LEAP admin dashboard. It provides a comprehensive view of all registered team members, including their names, emails, passwords, roles, and current statuses. The table also features action buttons that allow the admin to easily change user information, temporarily deactivate an account, or permanently remove a member, ensuring access control and system security remain in check. The "Add Member" option in the upper right corner also allows you to create new accounts for employees who have been assigned to new roles. This feature ensures that only authorized and active users can access the system, ensuring accountability, data integrity, and structured teamwork in managing the SMART LEAP program.



CITY GOVERNMENT OF BUTUAN
**Project Officer Console**

- Beneficiaries & Cases
- Application Check
- Repayments
- Training Pipeline

Logged in as
**Project Officer - po@smartleap.gov**

## Welcome back, Project Officer

Oversee beneficiary journeys, reconcile repayments, and coordinate the SMART LEAP training pipeline.

Refresh | Add beneficiary | Logout

| Total beneficiaries | Active cases | Closed cases | Pending applications |
|---|---|---|---|
| 2 | 0 | 0 | 1 |
| Profiles on record | Marked as Ongoing | Resolved files | Awaiting review |

**Beneficiary roster**

Search beneficiary or case number

| # | BENEFICIARY | CASE NO. | STATUS | SUPPORT REQUESTED | ACTIONS |
|---|---|---|---|---|---|
| | | | No beneficiaries recorded yet. | | |

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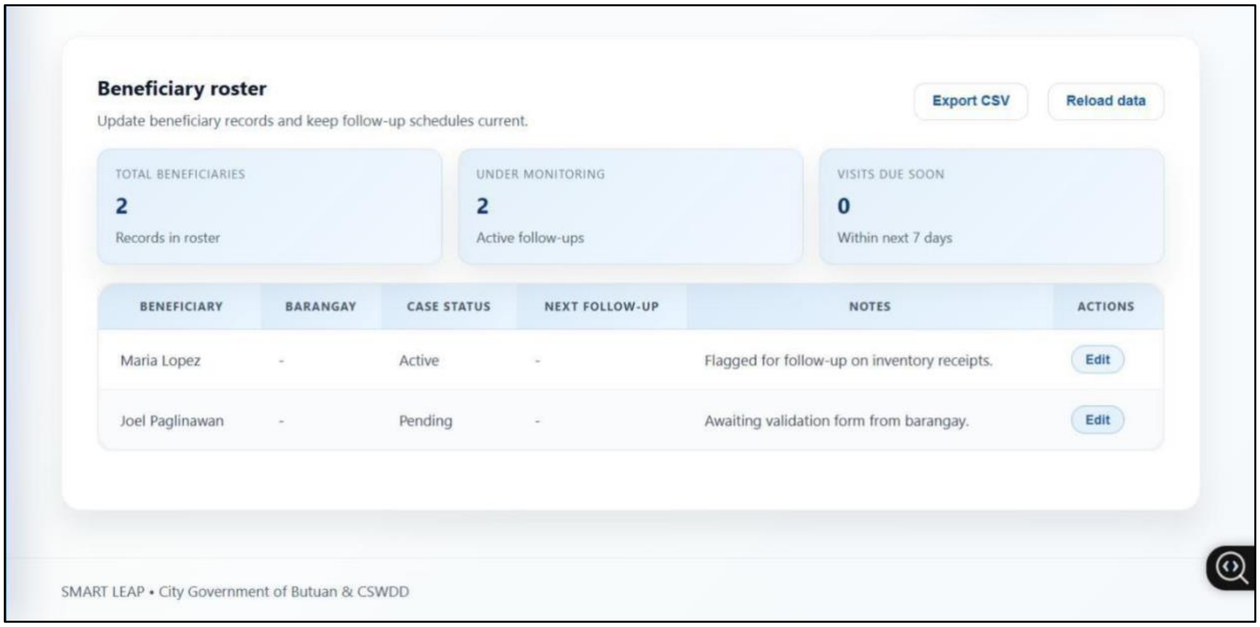
*Figure 21. PDO Dashboard*

Figure 21 illustrates that the SMART LEAP Project Development Officer (PDO) Dashboard is a role-specific interface designed to help PDOs effectively track beneficiary cases, address support inquiries, and plan program activities with clarity and efficiency. The beneficiary roster portion enables PDOs to look up individuals or case numbers, making it easy to locate and address specific instances. Each case is like a structured request for help, keeping track of the Beneficiary's progress through the application, training, help, and reimbursement. The sidebar of the dashboard features quick-access modules, including Beneficiaries & Cases, Application Check, Repayments, and Training Pipeline.



**Beneficiary roster**

Update beneficiary records and keep follow-up schedules current.

| TOTAL BENEFICIARIES | UNDER MONITORING | VISITS DUE SOON |
|---|---|---|
| 2 | 2 | 0 |
| Records in roster | Active follow-ups | Within next 7 days |

| BENEFICIARY | BARANGAY | CASE STATUS | NEXT FOLLOW-UP | NOTES | ACTIONS |
|---|---|---|---|---|---|
| Maria Lopez | - | Active | - | Flagged for follow-up on inventory receipts. | Edit |
| Joel Paglinawan | - | Pending | - | Awaiting validation form from barangay. | Edit |

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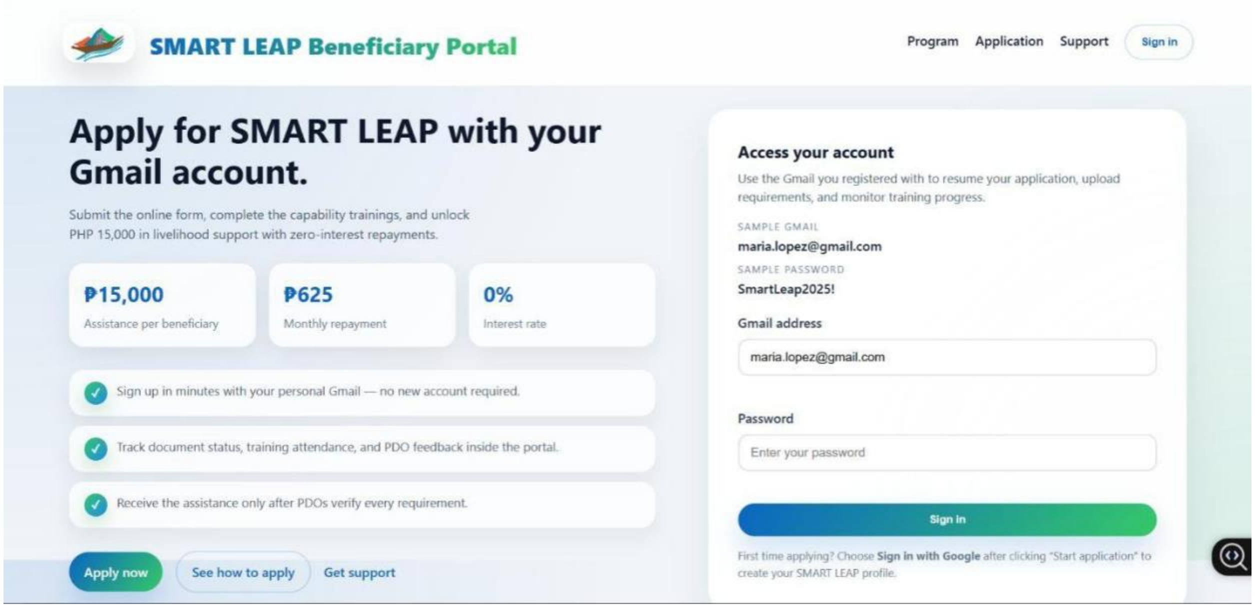
*Figure 22. Social Worker Dashboard*

Figure 22 illustrates that this dashboard displays the number of beneficiaries under active monitoring, the social worker's roster, and any scheduled visits in real-time. Social workers can also create reports for coordination meetings, compliance paperwork, or submission to program administrators using the dashboard's "Export CSV" feature. Encouraging transparency in communication helps ensure that field data is effectively shared across teams and adequately documented.



## Apply for SMART LEAP with your Gmail account.

Submit the online form, complete the capability trainings, and unlock PHP 15,000 in livelihood support with zero-interest repayments.

| | | |
|---|---|---|
| ₱15,000<br>Assistance per beneficiary | ₱625<br>Monthly repayment | 0%<br>Interest rate |

- ✓ Sign up in minutes with your personal Gmail — no new account required.
- ✓ Track document status, training attendance, and PDO feedback inside the portal.
- ✓ Receive the assistance only after PDOs verify every requirement.

**Access your account**

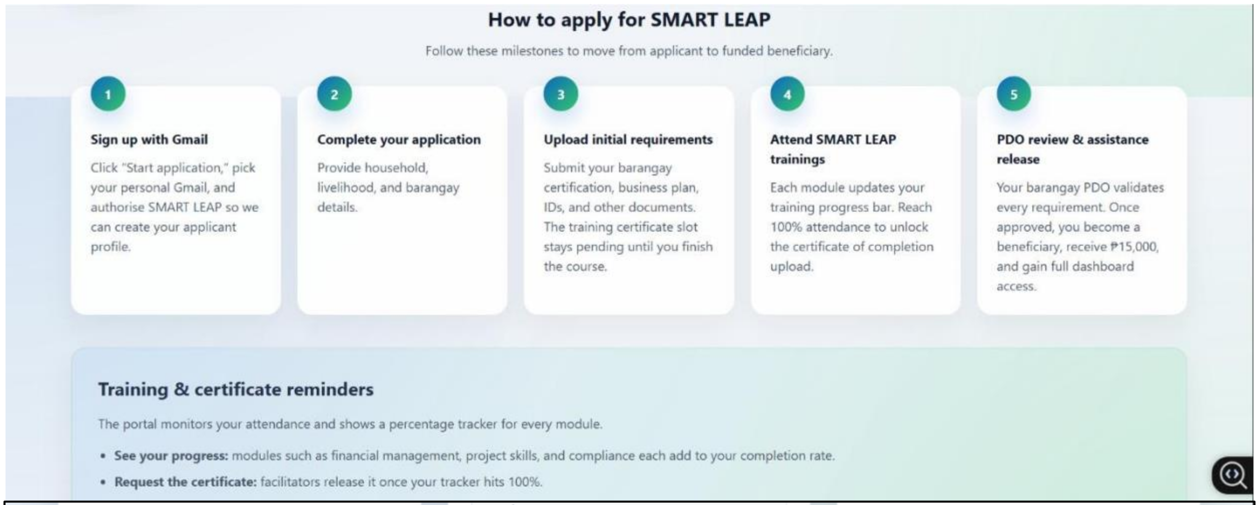
Use the Gmail you registered with to resume your application, upload requirements, and monitor training progress.

SAMPLE GMAIL
maria.lopez@gmail.com
SAMPLE PASSWORD
SmartLeap2025!

Gmail address: maria.lopez@gmail.com

Password: Enter your password

First time applying? Choose **Sign in with Google** after clicking "Start application" to create your SMART LEAP profile.



**How to apply for SMART LEAP**

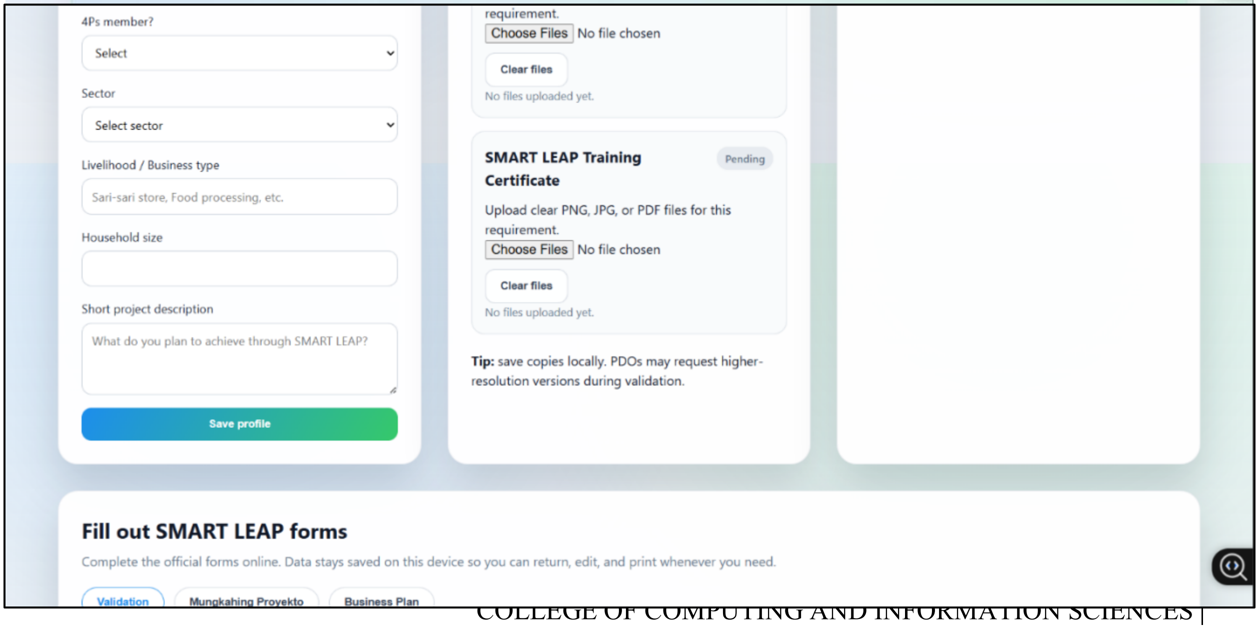
Follow these milestones to move from applicant to funded beneficiary.

1. **Sign up with Gmail** — Click "Start application," pick your personal Gmail, and authorise SMART LEAP so we can create your applicant profile.
2. **Complete your application** — Provide household, livelihood, and barangay details.
3. **Upload initial requirements** — Submit your barangay certification, business plan, IDs, and other documents. The training certificate slot stays pending until you finish the course.
4. **Attend SMART LEAP trainings** — Each module updates your training progress bar. Reach 100% attendance to unlock the certificate of completion upload.
5. **PDO review & assistance release** — Your barangay PDO validates every requirement. Once approved, you become a beneficiary, receive ₱15,000, and gain full dashboard access.

**Training & certificate reminders**

The portal monitors your attendance and shows a percentage tracker for every module.

- **See your progress:** modules such as financial management, project skills, and compliance each add to your completion rate.
- **Request the certificate:** facilitators release it once your tracker hits 100%.



4Ps member? Select

Sector: Select sector

Livelihood / Business type: Sari-sari store, Food processing, etc.

Household size:

Short project description: What do you plan to achieve through SMART LEAP?

requirement.
Choose Files No file chosen

No files uploaded yet.

**SMART LEAP Training Certificate** Pending

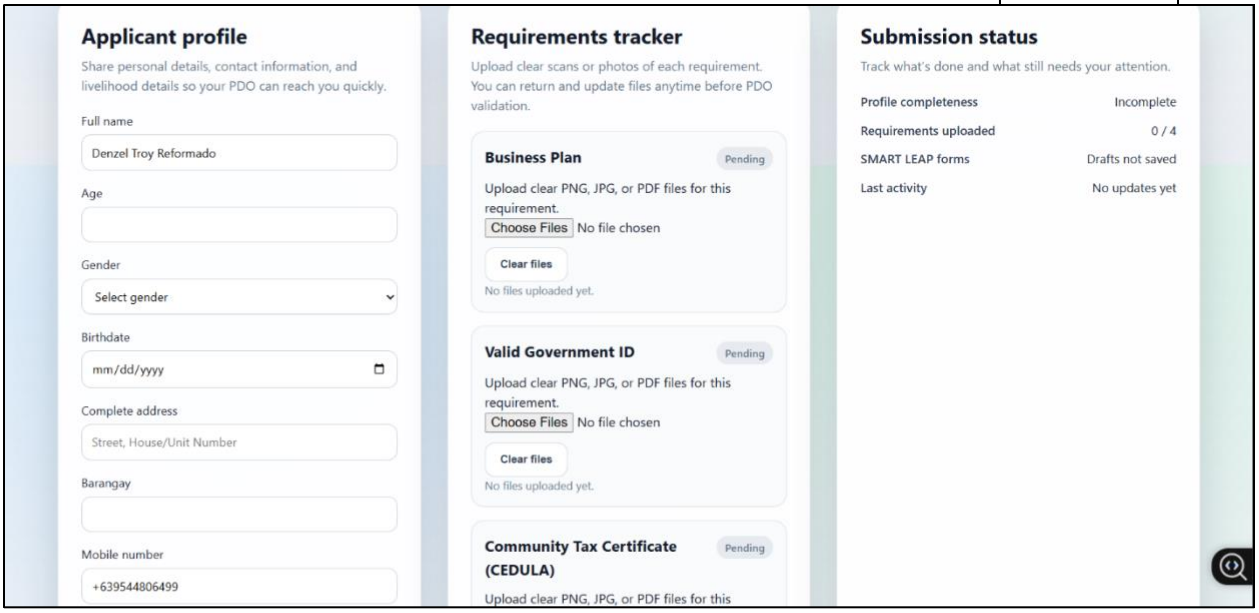
Upload clear PNG, JPG, or PDF files for this requirement.
Choose Files No file chosen

No files uploaded yet.

**Tip:** save copies locally. PDOs may request higher-resolution versions during validation.

**Fill out SMART LEAP forms**

Complete the official forms online. Data stays saved on this device so you can return, edit, and print whenever you need.



**Applicant profile**

Share personal details, contact information, and livelihood details so your PDO can reach you quickly.

Full name: Denzel Troy Reformado

Age:

Gender: Select gender

Birthdate: mm/dd/yyyy

Complete address: Street, House/Unit Number

Barangay:

Mobile number: +639544806499

**Requirements tracker**

Upload clear scans or photos of each requirement. You can return and update files anytime before PDO validation.

**Business Plan** — Pending

Upload clear PNG, JPG, or PDF files for this requirement.

Choose Files No file chosen

Clear files

No files uploaded yet.

**Valid Government ID** — Pending

Upload clear PNG, JPG, or PDF files for this requirement.

Choose Files No file chosen

Clear files

No files uploaded yet.

**Community Tax Certificate (CEDULA)** — Pending

Upload clear PNG, JPG, or PDF files for this

**Submission status**

Track what's done and what still needs your attention.

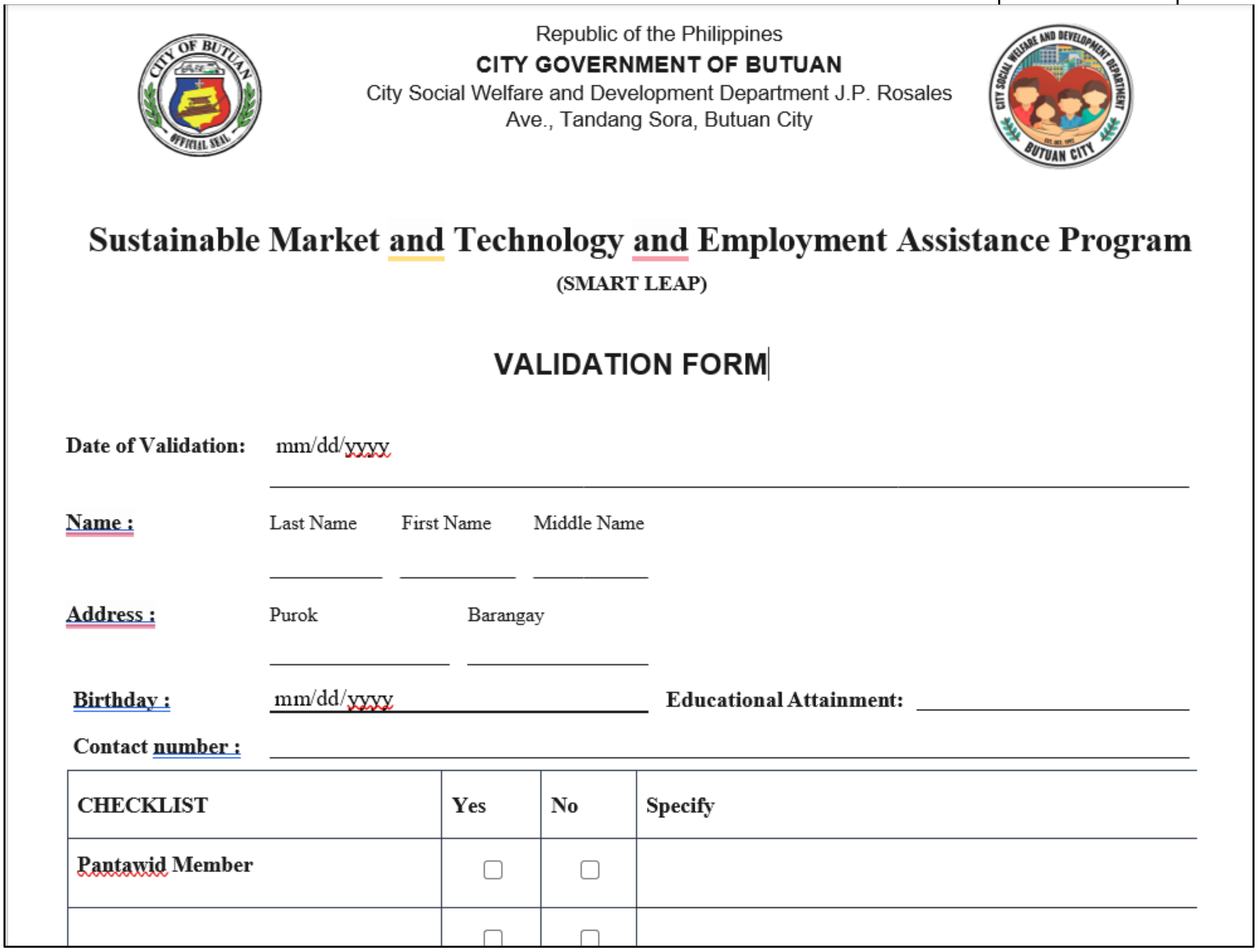
| | |
|---|---|
| Profile completeness | Incomplete |
| Requirements uploaded | 0 / 4 |
| SMART LEAP forms | Drafts not saved |
| Last activity | No updates yet |

*Figure 23. Beneficiary Portal*

Figure 23 illustrates the process, starting with the Beneficiary Portal, where individuals can apply for livelihood support using their Gmail accounts. The SMART LEAP system provides a simplified online experience for program personnel and applicants. This portal explains the program's advantages and streamlines access by enabling users to log in without creating a new account. The program offers financial aid of ₱15,000 with no interest and a modest monthly repayment of ₱625. Candidates follow a procedure that includes filling out online forms, uploading requirements, attending capability training, and monitoring their progress throughout the portal. The interface also incorporates login areas for Gmail and a password to reinforce safe and customized access.

After entering the system, candidates complete a comprehensive SMART LEAP application form that asks for demographic and household data. Full name, phone number, age, residence, barangay, and household size are among the fields; these details help program staff determine eligibility and set priorities for assistance. To ensure that the data gathered aligns with operational processes and verification protocols, we designed this form to reflect the forms used by Program Development Officers (PDOs) and Social Workers during validation. The form's format encourages completeness and clarity, which reduces errors and speeds up processing.

Users enter socioeconomic data and livelihood details later in the application. These consist of business type, years of operation, sector classification, gender, and 4Ps membership status. Whether the applicant is operating a sari-sari store or working informally, this area is essential for customizing help to their situation. In addition to standardizing data for backend analysis, dropdown menus and text fields make the form easy to use.



Republic of the Philippines
**CITY GOVERNMENT OF BUTUAN**
City Social Welfare and Development Department J.P. Rosales Ave., Tandang Sora, Butuan City

## Sustainable Market and Technology and Employment Assistance Program

**(SMART LEAP)**

## VALIDATION FORM

**Date of Validation:** mm/dd/yyyy ______________________________

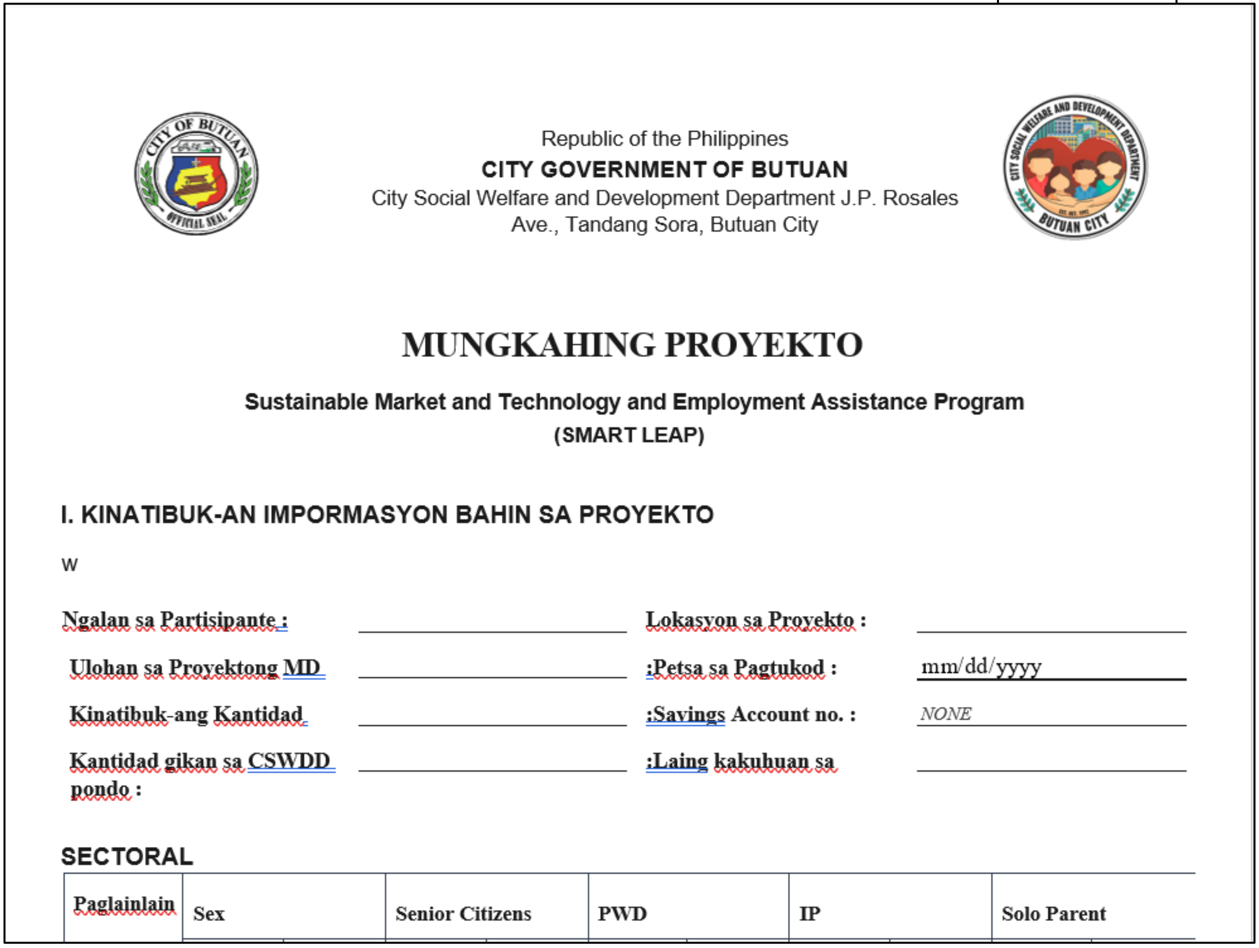
**Name :** Last Name __________ First Name __________ Middle Name __________

**Address :** Purok ______________ Barangay ______________

**Birthday :** mm/dd/yyyy ______________ **Educational Attainment:** ______________________

**Contact number :** ______________________________

| **CHECKLIST** | **Yes** | **No** | **Specify** |
|---|---|---|---|
| **Pantawid Member** | ☐ | ☐ | |
| | ☐ | ☐ | |



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**CITY GOVERNMENT OF BUTUAN**
City Social Welfare and Development Department J.P. Rosales Ave., Tandang Sora, Butuan City

## MUNGKAHING PROYEKTO

**Sustainable Market and Technology and Employment Assistance Program**
**(SMART LEAP)**

### I. KINATIBUK-AN IMPORMASYON BAHIN SA PROYEKTO

w

**Ngalan sa Partisipante :** ______________ **Lokasyon sa Proyekto :** ______________

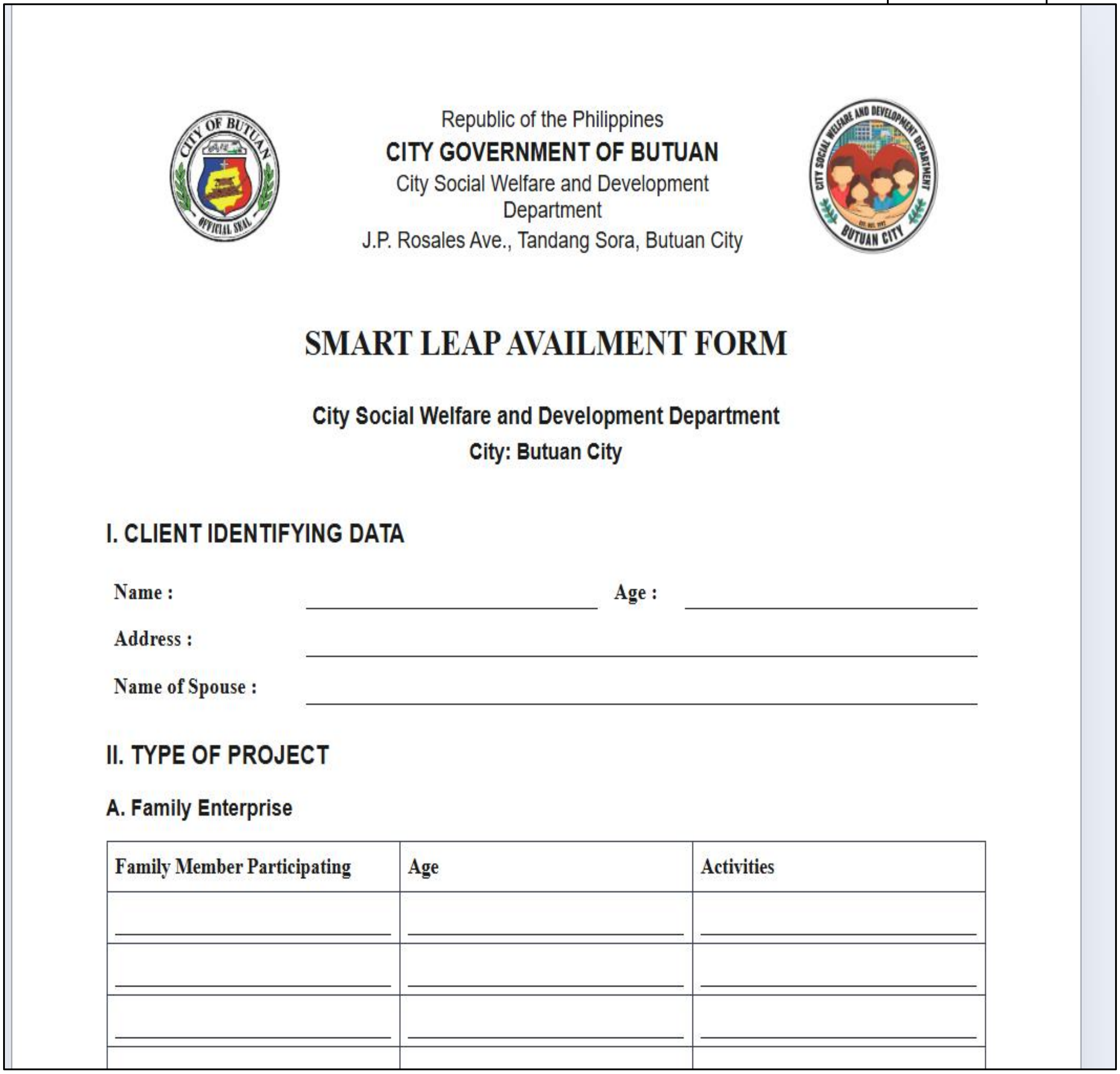
**Ulohan sa Proyektong MD** ______________ **:Petsa sa Pagtukod :** mm/dd/yyyy

**Kinatibuk-ang Kantidad** ______________ **:Savings Account no. :** *NONE*

**Kantidad gikan sa CSWDD pondo :** ______________ **:Laing kakuhuan sa** ______________

### SECTORAL

| Paglainlain | Sex | Senior Citizens | PWD | IP | Solo Parent |
|---|---|---|---|---|---|



Republic of the Philippines
**CITY GOVERNMENT OF BUTUAN**
City Social Welfare and Development Department
J.P. Rosales Ave., Tandang Sora, Butuan City

# SMART LEAP AVAILMENT FORM

**City Social Welfare and Development Department**
**City: Butuan City**

## I. CLIENT IDENTIFYING DATA

**Name :** ______________________ **Age :** ______________________

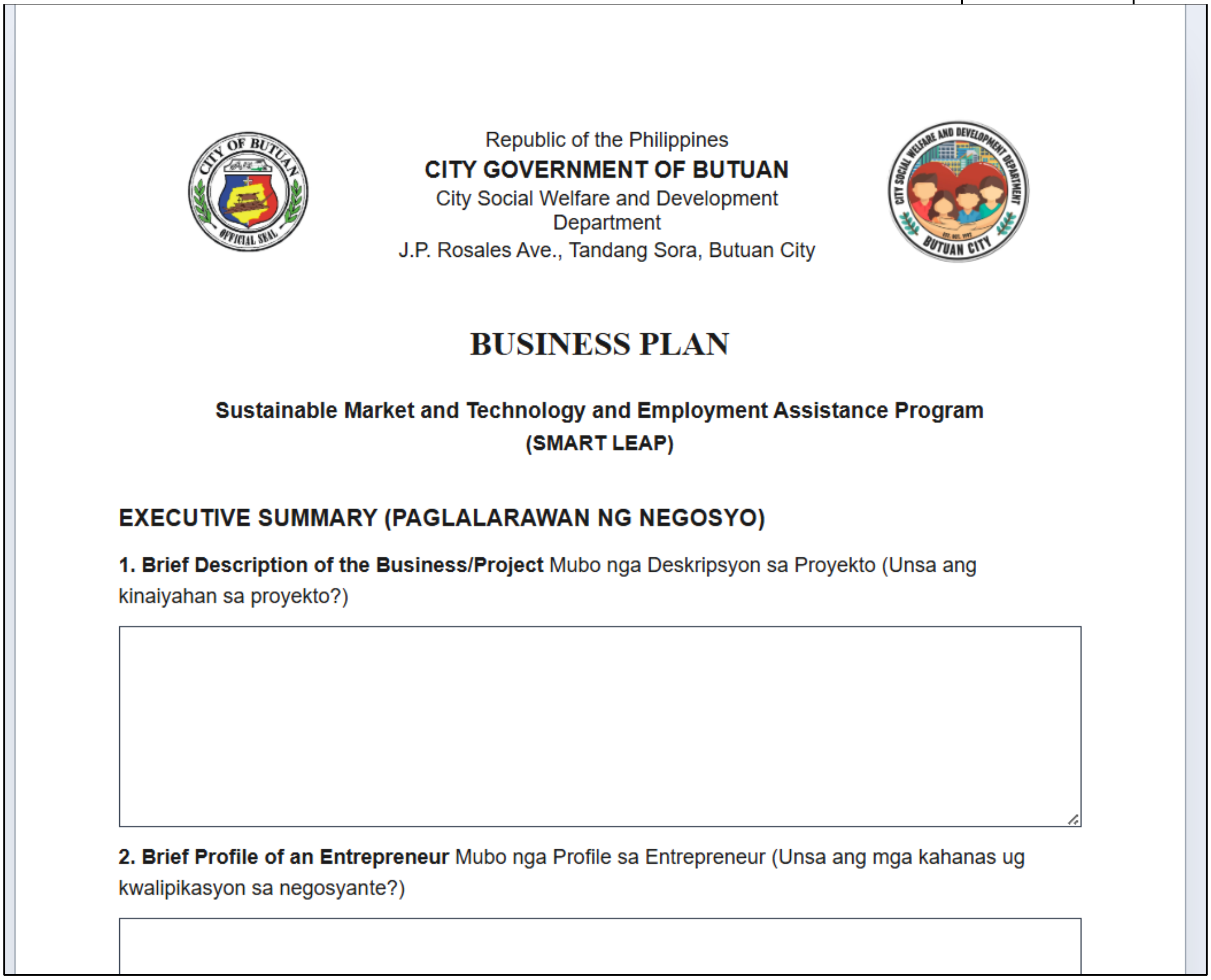
**Address :** ____________________________________________

**Name of Spouse :** ____________________________________________

## II. TYPE OF PROJECT

### A. Family Enterprise

| **Family Member Participating** | **Age** | **Activities** |
|---|---|---|
| | | |
| | | |
| | | |
| | | |



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**CITY GOVERNMENT OF BUTUAN**
City Social Welfare and Development Department
J.P. Rosales Ave., Tandang Sora, Butuan City

**BUSINESS PLAN**

**Sustainable Market and Technology and Employment Assistance Program (SMART LEAP)**

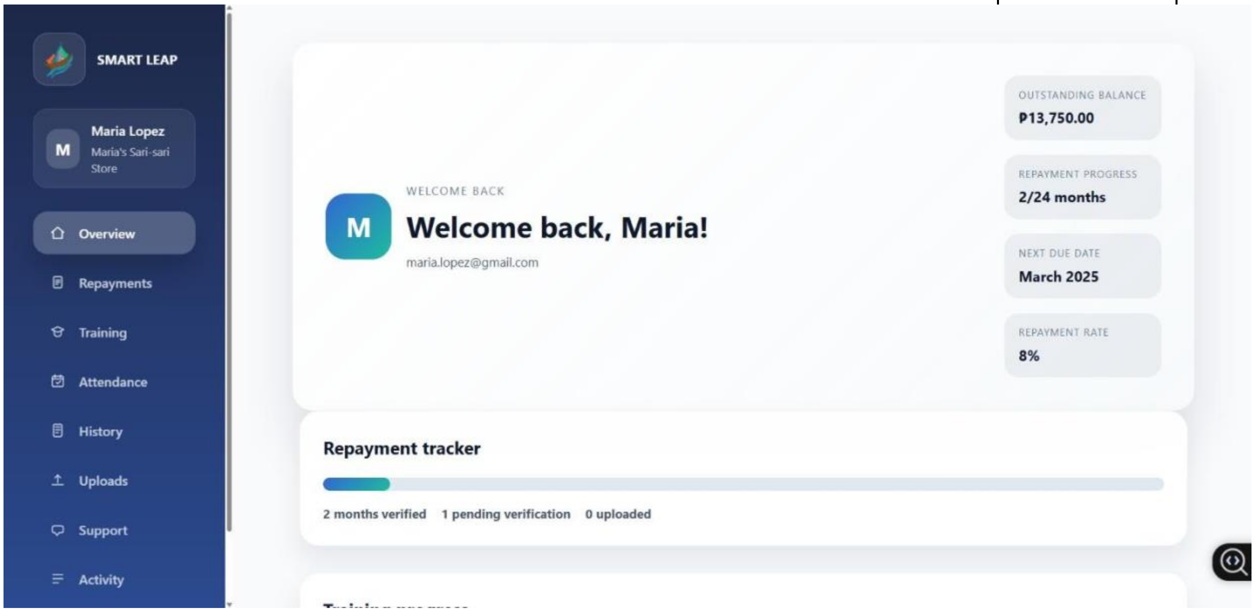
**EXECUTIVE SUMMARY (PAGLALARAWAN NG NEGOSYO)**

**1. Brief Description of the Business/Project** Mubo nga Deskripsyon sa Proyekto (Unsa ang kinaiyahan sa proyekto?)

**2. Brief Profile of an Entrepreneur** Mubo nga Profile sa Entrepreneur (Unsa ang mga kahanas ug kwalipikasyon sa negosyante?)

*Figure 24. Beneficiary Application Forms*

Figure 24 shows the full set of online forms that people who want to apply for the SMART LEAP (Sustainable Market and Technology and Employment Assistance Program) must complete as part of their application for financial help and a project idea. These are the Availment form, the SMART LEAP Validation form, the Mungkahing Proyekto form, and the Business Plan. They are all necessary to determine whether the suggested livelihood project is eligible, ready, and feasible.


SMART LEAP
Overview
Repayments
Training
Attendance
History
Uploads
Support
Activity

Maria Lopez
Maria's Sari-sari Store

WELCOME BACK

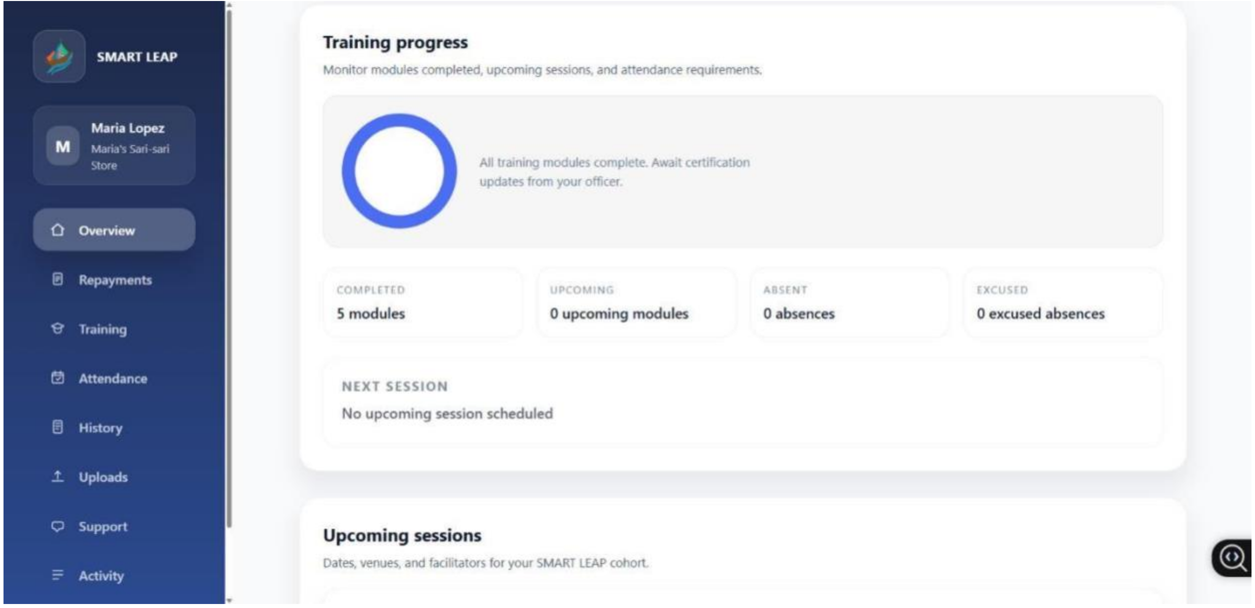
**Welcome back, Maria!**

maria.lopez@gmail.com

OUTSTANDING BALANCE: ₱13,750.00
REPAYMENT PROGRESS: 2/24 months
NEXT DUE DATE: March 2025
REPAYMENT RATE: 8%

**Repayment tracker**

2 months verified  1 pending verification  0 uploaded


SMART LEAP
Overview
Repayments
Training
Attendance
History
Uploads
Support
Activity

Maria Lopez
Maria's Sari-sari Store

**Training progress**

Monitor modules completed, upcoming sessions, and attendance requirements.

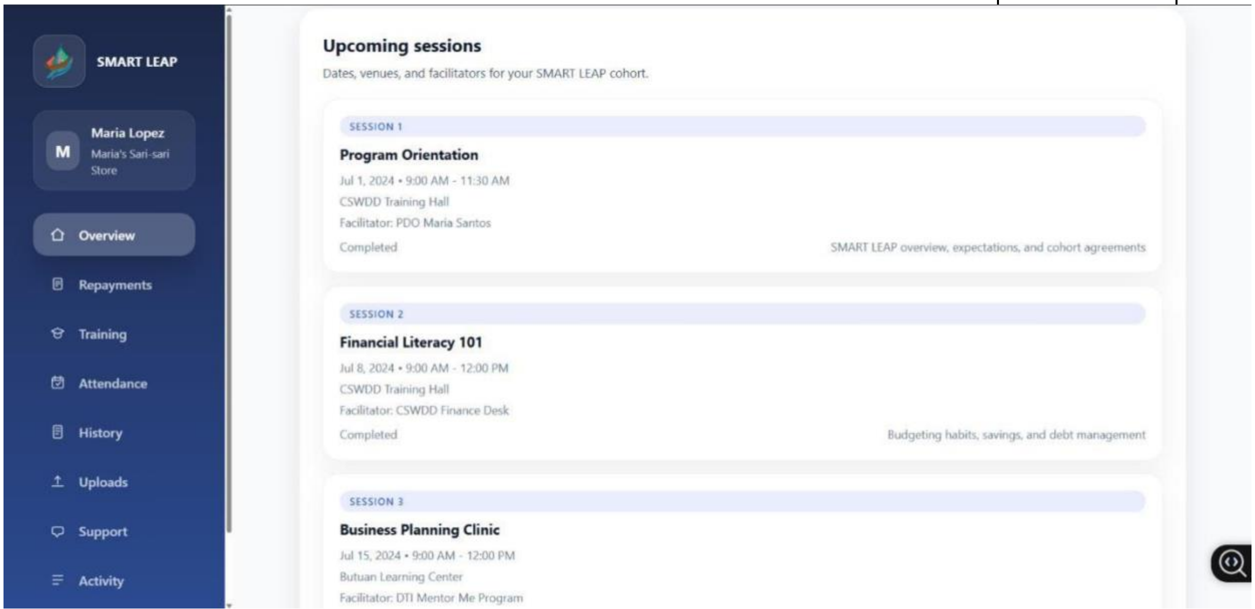
All training modules complete. Await certification updates from your officer.

| COMPLETED | UPCOMING | ABSENT | EXCUSED |
|---|---|---|---|
| 5 modules | 0 upcoming modules | 0 absences | 0 excused absences |

NEXT SESSION
No upcoming session scheduled

**Upcoming sessions**

Dates, venues, and facilitators for your SMART LEAP cohort.


SMART LEAP
M
Maria Lopez
Maria's Sari-sari Store
Overview
Repayments
Training
Attendance
History
Uploads
Support
Activity

**Upcoming sessions**

Dates, venues, and facilitators for your SMART LEAP cohort.

SESSION 1

**Program Orientation**

Jul 1, 2024 • 9:00 AM - 11:30 AM
CSWDD Training Hall
Facilitator: PDO Maria Santos
Completed — SMART LEAP overview, expectations, and cohort agreements

SESSION 2

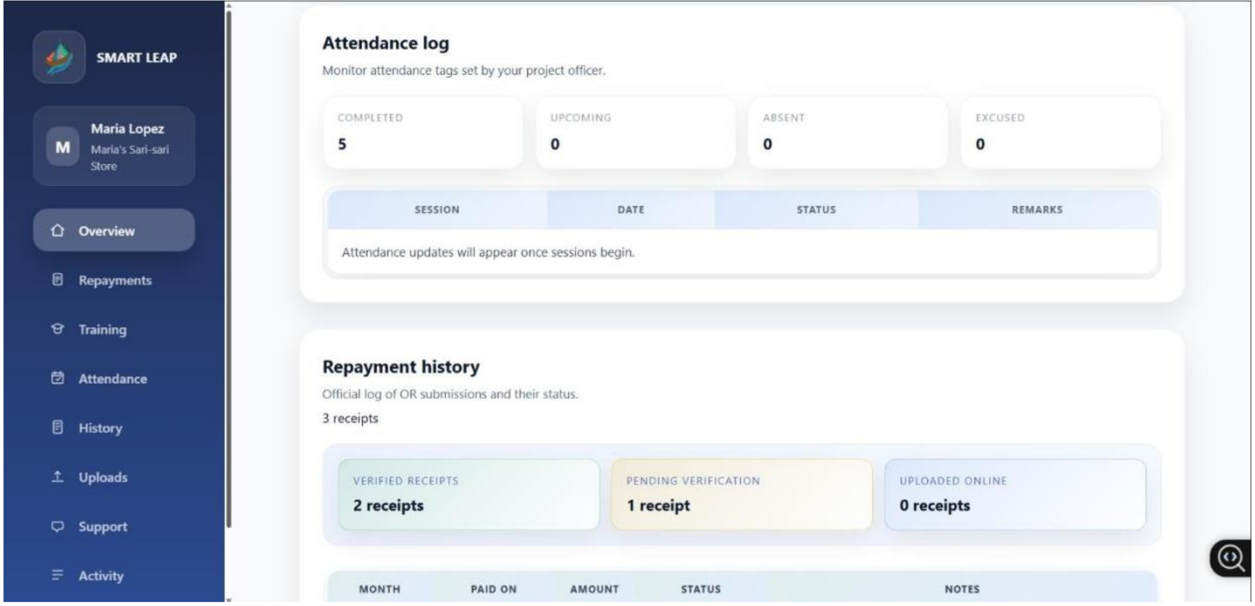
**Financial Literacy 101**

Jul 8, 2024 • 9:00 AM - 12:00 PM
CSWDD Training Hall
Facilitator: CSWDD Finance Desk
Completed — Budgeting habits, savings, and debt management

SESSION 3

**Business Planning Clinic**

Jul 15, 2024 • 9:00 AM - 12:00 PM
Butuan Learning Center
Facilitator: DTI Mentor Me Program


SMART LEAP
M
Maria Lopez
Maria's Sari-sari Store
Overview
Repayments
Training
Attendance
History
Uploads
Support
Activity

**Attendance log**

Monitor attendance tags set by your project officer.

| COMPLETED | UPCOMING | ABSENT | EXCUSED |
|---|---|---|---|
| 5 | 0 | 0 | 0 |

| SESSION | DATE | STATUS | REMARKS |
|---|---|---|---|
| Attendance updates will appear once sessions begin. | | | |

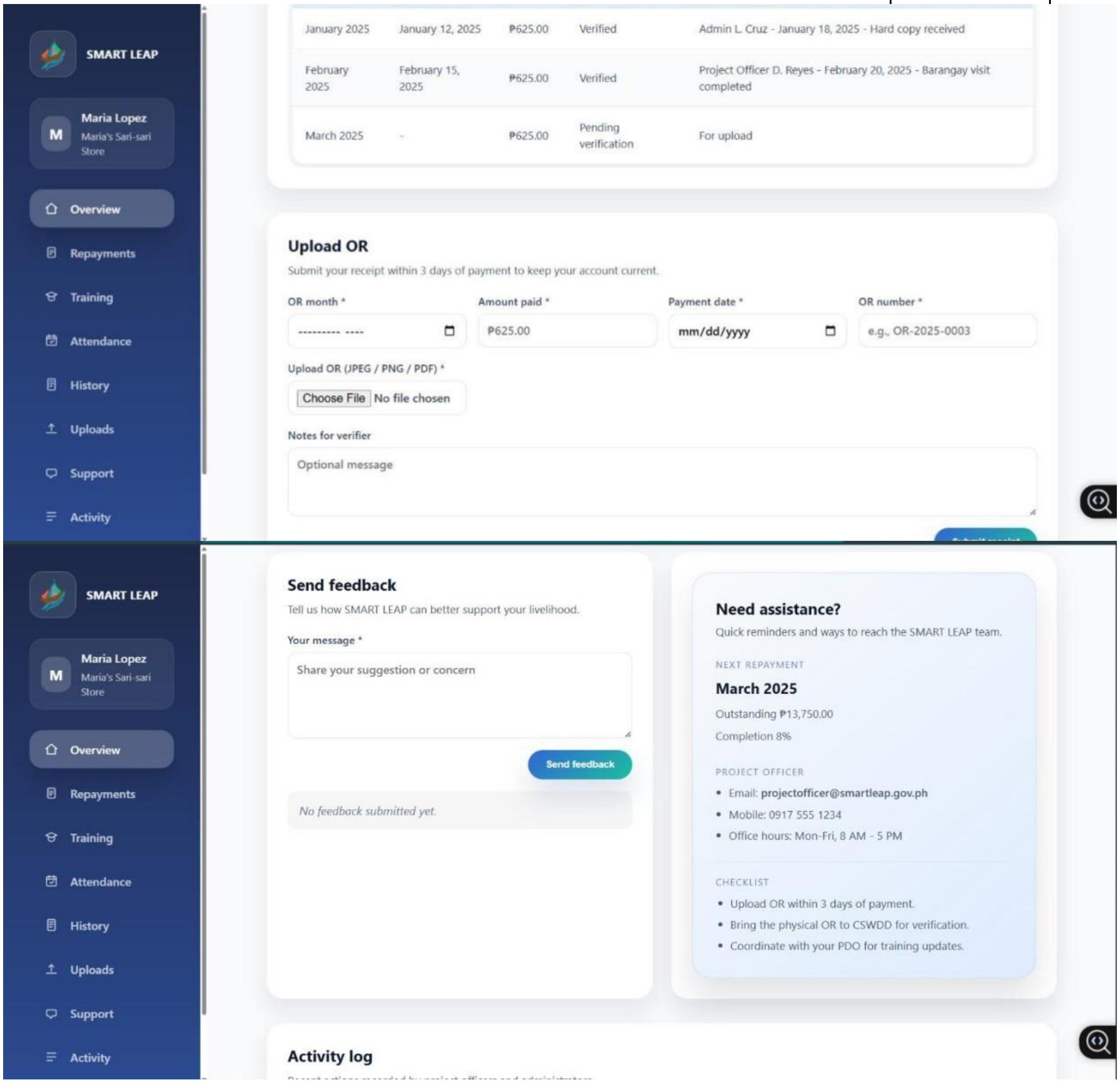
**Repayment history**

Official log of OR submissions and their status.

3 receipts

| VERIFIED RECEIPTS | PENDING VERIFICATION | UPLOADED ONLINE |
|---|---|---|
| **2 receipts** | **1 receipt** | **0 receipts** |

| MONTH | PAID ON | AMOUNT | STATUS | NOTES |
|---|---|---|---|---|



| | | | | |
|---|---|---|---|---|
| January 2025 | January 12, 2025 | ₱625.00 | Verified | Admin L. Cruz - January 18, 2025 - Hard copy received |
| February 2025 | February 15, 2025 | ₱625.00 | Verified | Project Officer D. Reyes - February 20, 2025 - Barangay visit completed |
| March 2025 | - | ₱625.00 | Pending verification | For upload |

**Upload OR**

Submit your receipt within 3 days of payment to keep your account current.

OR month * ---------- ----
Amount paid * ₱625.00
Payment date * mm/dd/yyyy
OR number * e.g., OR-2025-0003
Upload OR (JPEG / PNG / PDF) * Choose File No file chosen
Notes for verifier Optional message

**Send feedback**

Tell us how SMART LEAP can better support your livelihood.

Your message * Share your suggestion or concern

Send feedback

*No feedback submitted yet.*

**Need assistance?**

Quick reminders and ways to reach the SMART LEAP team.

NEXT REPAYMENT

**March 2025**

Outstanding ₱13,750.00

Completion 8%

PROJECT OFFICER

- Email: projectofficer@smartleap.gov.ph
- Mobile: 0917 555 1234
- Office hours: Mon-Fri, 8 AM - 5 PM

CHECKLIST

- Upload OR within 3 days of payment.
- Bring the physical OR to CSWDD for verification.
- Coordinate with your PDO for training updates.

**Activity log**

*Figure 25. Beneficiary Dashboard*

Figure 24 illustrates that the SMART LEAP Beneficiary Portal is a secure and user-friendly portal that empowers applicants to monitor their livelihood support journey with ease and transparency. It starts with a simplified Gmail account login procedure that does not require new login information. After logging in, recipients can apply, complete the necessary training, and receive ₱15,000 in financial aid with interest-free repayment terms. The portal explains the program's structure in detail, including the ₱625 monthly repayments and the requirement that aid be given upon the completion of an earlier Program Officer's (POO) early payback. This onboarding UI enables users to track their progress independently and sets clear expectations. Personal, household, and livelihood data are gathered through organized parts of the application form.

For eligibility screening and demographic profiling, applicants are required to enter their full name, contact information, age, address, barangay, and household size. Gender, 4Ps membership, sector classification, firm type, and years of operation are captured via additional fields. Whether an applicant is working informally or owns a sari-sari business, these specifics assist in customizing support to their situation. The form ensures alignment between frontend input and backend verification by using text fields and dropdowns, providing user convenience and reflecting the types of information that Program Development Officers (PDOs) and Social Workers examine during validation. Beneficiaries can track their training and reimbursement progress on a personalized dashboard after enrollment. The training module displays completed sessions, attendance logs, and certification status.

For example, Maria Lopez's dashboard shows five completed modules, including Program Orientation, Financial Literacy, and Business Planning, that have no upcoming sessions or absences. Beneficiaries are kept informed and held accountable through this structured approach. Verified payments, pending receipts, and outstanding sums are displayed in the repayment area. Using a specific form, users can upload official receipts (ORs) by entering the OR number, month, amount, and date of payment. They can also upload a file and include notes for verifiers. This ensures that repayments are accurately tracked and recorded. Finally, SMART LEAP's dedication to providing prompt service is further evidenced by the portal's help and feedback section. Recipients can use a feedback form to make suggestions or grievances instantly. They may also be reminded of impending obligations, such as making business permit applications or working with their PDO. Project officers' email addresses, phone numbers, and office hours are all prominently displayed. To ensure that beneficiaries feel supported and heard throughout their journey, this area bridges digital access with human assistance.

**Table 1**

**Software Platforms, Development Environment and Tools**

| Components | Specification | Usage |
|---|---|---|
| Internet Browser | Any (Chrome, Edge, Firefox, etc.) | Users can access and view information from the internet with a web browser. After a successful login, the SMART LEAP's interface loads, allowing users to access its functionalities. |
| Front-end | HTML | HTML specifies the structural elements of the Web-Based SMART LEAP System's interface, including forms, text, and other critical components. |
| | Tailwind CSS | CSS framework that allows developers to design custom, modern-looking interfaces quickly. |
| | JavaScript | To make the website more interactive and provide users with more functionality, JavaScript handles all client-side functionality. |
| Backend | MySQL (Database) | Handles structured data storage and management. Through phpMyAdmin, developers can create tables, run queries, and maintain database integrity for the system. |
| | PHP | PHP is used on the server side to manage interactions between user input and the database, ensuring smooth data flow. |

| | | |
|---|---|---|
| Server | Apache | Using the XAMPP control panel, developers can manage the behavior of its components, ensuring smooth, easy access to local host servers such as Apache and MySQL. |

**Table 2**

| **Components** | **Specification** | **Usage** |
|---|---|---|
| Device | RAM (8 GB) | This is like the computer's short-term memory. It helps the computer run many things at once and makes apps and files open faster. |
| | Processor (Intel Core i5) | The main component of the computer—the processor, or CPU—is responsible for executing commands from the website. It also makes data processing easier. |
| | SSD (128 GB) | This is where the computer saves programs, reports, and files. It works faster than old hard drives, so you can open and save things quickly. |
| | Flash Storage (8 GB) | These storage solutions ensure quick retrieval and secure long-term data handling. |
| Printer | Any Ink Jet Printer Units | Inkjet printers produce clear, high-quality output used to print compliance documentation, data summaries, and audit |

| | | |
|---|---|---|
| | | reports. Since these printers are reasonably priced and multipurpose, they can be used for routine audit management printing, including creating formal documents for internal use or sending them to stakeholders. |

**Hardware Requirements**

## System Evaluation

The evaluation of the Web-Based SMART LEAP System was conducted using two approaches: the ISO 25010 quality model. This dual framework ensures that the System is not only functional, efficient, and usable but also that its AI component delivers accurate and reliable recommendations. *Data was analyzed and interpreted based on the following parameters:*

**Table 3**

| Adjectival Rating | Scale Range (Mean) | Verbal Interpretation |
|---|---|---|
| 4 | 3.50 – 4.0 | Very Functional |
| 3 | 2.50 – 3.49 | Functional |
| 2 | 1.50 – 2.49 | Moderately Functional |

| | | |
|---|---|---|
| 1 | 1.0 – 1.49 | Poor Functional |

**Functional Suitability**

Table 3 presents the criteria used to evaluate a product or System's functional suitability according to the ISO survey.

This measurement highlights the degree to which the System's modules—such as reports and AI analysis—align with user expectations and deliver the intended outcomes.

**Table 4**

**Performance Efficiency**

| Adjectival Rating | Scale Range (Mean) | Verbal Interpretation |
|---|---|---|
| 4 | 3.50 – 4.0 | Very Efficient |
| 3 | 2.50 – 3.49 | Efficient |
| 2 | 1.50 – 2.49 | Moderately Efficient |
| 1 | 1.0 – 1.49 | Poor Efficient |

Table 4 outlines the scoring parameters used to assess the System's overall performance efficiency in executing its functions and tasks.

**Table 5**

**Usability**

| **Adjectival Rating** | **Scale Range (Mean)** | **Verbal Interpretation** |
|---|---|---|
| 4 | 3.50 – 4.0 | Very Usable |
| 3 | 2.50 – 3.49 | Usable |
| 2 | 1.50 – 2.49 | Moderately Usable |
| 1 | 1.0 – 1.49 | Poor Usable |

Table 5 presents the scoring parameters used to assess the System's usability.

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## Research Ethical Standard

### A. Protection of Intellectual Property Rights (IPR)

The SMART LEAP system encompasses the development of new technologies and procedures, including digital remittance uploads, a repayment verification mechanism, and a performance-tracking module. These components may be eligible for intellectual property protection under Republic Act 8293, the Intellectual Property Code of the Philippines. The program's source codes, database structures, and interface designs were developed by the research team under Saint Michael College of Caraga (SMCC), therefore, these works are owned by SMCC and protected by copyright law. Additionally, system-generated outputs such as user manuals, reports, and training materials form part of the institution's intellectual property. To safeguard these creations, the research team will pursue the necessary copyright registration, branding, and potential trademark or patent applications for the system name and logo. The team also clarifies that SMART LEAP uses third-party software and frameworks (e.g., PHP, MySQL, Bootstrap, CSS) licensed under open-source agreements. These external tools remain the property of their respective developers, and the project's authorship applies only to original code and documentation produced for the system, in full compliance with applicable open-source licensing terms.

### B. Informed Consent

Everyone who took part in the system evaluation, beneficiaries, social workers, project development officers (PDOs), and admin, was informed about the study's objectives, process, methods, potential risks, and benefits. Written informed consent was secured from all CSWDD personnel before prototype testing. The consent form was provided in simple english to ensure comprehension, and participants were informed of their voluntary role and their right to withdraw from the study at any time without penalty. They were allowed to ask questions and voluntarily agreed to the terms before signing, with no coercion, pressure, or compensation provided. No minors, animals, or biological testing were involved in this study; all human participants were treated ethically and in accordance with the research standards set by Saint Michael College of Caraga (SMCC) and the City Social Welfare and Development Department (CSWDD), Butuan City.

### C. Data Privacy and Confidentiality

Since the SMART LEAP system stores sensitive information about beneficiaries such as personal details, repayment records, payments, and demographic profiles-strong data protection measures were implemented. All collected data were securely stored in password-protected computers and encrypted databases, with access strictly limited to authorized researchers and system administrators. The project complies with all required legal data protection requirements, including the Data Privacy Act of 2012 (Republic Act No. 10173), institutional data security policies, and CSWDD guidelines on handling personal and financial information. To ensure anonymity during analysis, names were replaced with unique ID codes. For verification purposes, data will be retained for five (5) years before being permanently deleted or anonymized.

### D. Voluntary Participation and Freedom to Withdraw

Participation in this study was entirely voluntary. All respondents and system testers joined of their own free will and were clearly informed that they could withdraw from the research at any point without any penalties, loss of benefits, or impact on their employment evaluation. This was explained both verbally and in writing before participation. If a participant decided to withdraw, all related data were immediately removed from the study records. Participation was not tied to employment evaluation or program benefits. This practice aligns with the ethical principles of autonomy and respect for human dignity.

### E. Minimization of Harm and Risk Management

The study was carefully designed to ensure the safety and comfort of all participants throughout all activities. There were no intrusive, deceptive, or emotionally distressing procedures, and participants were assured that no psychological, professional, or financial harm would result from their involvement in the testing. The system demonstration and usability testing were conducted in a supportive environment with proper orientation and technical assistance. No physical, emotional, psychological, or professional harm occurred during participation. The process ensured the safety and welfare of both participants and institutional staff involved in the study.

### F. Beneficence and Contribution to Knowledge

The SMART LEAP project offers direct social and professional benefits to CSWDD, its staff, and program beneficiaries by enhancing the management of livelihood assistance, promoting transparency, and improving repayment tracking and operational efficiency. The study's outcomes aim to support more efficient digital Government-to-Person (G2P) transactions, contributing to ethical justification by demonstrating tangible community and institutional benefits. The system also promotes digital literacy and financial responsibility among beneficiaries, empowering them to manage their livelihood assistance more effectively. Furthermore, the findings and the developed SMART LEAP system will be made available to CSWDD upon completion, supporting the field of e-governance and public information. management and serving as a model for other local government units implementing digitalized social protection programs.

### G. Justice and Fair Participation

All individuals had an equal opportunity to participate in the study regardless of gender, economic status, or residence. Participants were selected based on role relevance-administrators, PDOs, and social workers-without bias regarding gender, position, or employment status. The sampling procedure ensured balanced representation across departments and stakeholder groups, including beneficiaries, to reflect the program's diversity. No participant was excluded or mistreated. Transparent and ethical selection criteria were applied, guaranteeing inclusiveness and fairness in both participation and data handling.

### H. Data Integrity and Accuracy

The research team verified all SMART LEAP system data entries, logs, and records to maintain accuracy, reliability, and integrity, ensuring that no fabrication, bias, or manipulation occurred. Only validated and authe data ere used during the development and evaluation process. Data collection and evalua validated and cross-checked by the advisory panel. The System includes built-in validation mechanisms for official receipts, repayment tracking, and attendance records to prevent duplication or falsification. Digital tools, such as Grammarly and ChatGPT, were used for documentation and reporting purposes only, without affecting raw data integrity. Any limitations, errors, or discrepancies identified during the process were documented to maintain transparency and accuracy throughout the study.

### I. Transparency and Honesty in Reporting

All findings-whether favorable or unfavorable-will be reported truthfully, objectively, and without exaggeration. The research team does not fabricate or manipulate data. All references, figures, and system components were properly cited, and plagiarism-checking tools as well as Grammarly were used to maintain accuracy and integrity in reporting. The final research report will clearly present the System's limitations and challenges encountered during development and testing. Transparent reporting ensures accountability and enables future researchers and institutions to replicate, build upon, or improve this work with integrity.

### J. Use of Patented or Copyrighted Materials

SMART LEAP system was developed using third-party technologies and frameworks, including PHP, MySQL, Bootstrap, Tailwind CSS, HTML, and Draw.io, all of which are used under their respective open-source or educational licenses. All icons, templates, and external resources used in the interface were properly credited and cited, and full compliance with their licensing terms has been ensured. No proprietary or patented software was used without proper authorization. This approach ensures adherence to intellectual property laws and upholds respect for the rights of original creators.

### K. Ethical Considerations for Animals and Human Trials

This study involved only human participants in a non-invasive, digital testing. environment. Only adult CSWDD staff participated, and all data were gathered solely for usability and system evaluation purposes. No vulnerable populations, medical procedures, or invasive experiments were involved. All participants were adequately informed of their rights, responsibilities, and roles before using the System, and their confidentiality, fairness, and autonomy were fully respected. Before testing at CSWDD Butuan City, ethical clearance was obtained from the Saint Michael College of Caraga Research Ethics Committee, ensuring adherence to ethical standards for studies involving human subjects.

### L. Responsible Use of AI and Other Related Technologies

Although primarily web-based, the SMART LEAP system integrates algorithmic features for automated repayment validation and data visualization. The system's AI analytics feature was designed for performance recommendations only. No automated decision-making affecting beneficiaries' rights or welfare was implemented, and all AI-generated outputs were validated by human reviewers to ensure accuracy, fairness, and ethical integrity. Data used for testing were anonymized to avoid bias or misinterpretation. AI-assisted tools were also utilized to streamline the literature review and data analysis process, supporting efficiency while maintaining human verification, academic integrity, and responsible use of technology.

### M. Ethical Clearance and Institutional Approval Approval

Before conducting surveys, interviews, or system tests, the research team sought and received formal approval from the Research Ethics Committee of Saint Michael College of Caraga (SMCC) This capstone project received ethical clearance from the SMCC Research Ethics Committee in January 2026. The committee evaluated the project's objectives, data handling methods, and participant safeguards, and the approval specifically covered system testing, data collection, and prototype evaluation under the guidelines of the institution's Research Ethics Manual. This ensures that the SMART LEAP study complies fully with both institutional and national ethical research standards for studies involving human participants.

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| ORGANIZATION | POSITIONS HELD |
|---|---|
| CCS CLUB | TRESURER |

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| | |
|---|---|
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**Educational Attainment**

| | |
|---|---|
| **Elementary** | : Hinandayan Elementary School |
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**Membership/Affiliations**

| ORGANIZATION | POSITIONS HELD |
|---|---|
| NONE | NONE |

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| | |
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| | |
|---|---|
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| **Tertiary** | : Saint Michael College of Caraga |

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|---|---|
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### Membership/Affiliations

| ORGANIZATION | POSITIONS HELD |
|---|---|
| NONE | NONE |

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**Tertiary**: Saint Michael College of Caraga

**Membership/Affiliations**

| ORGANIZATION | POSITIONS HELD |
|---|---|
| NONE | NONE |